

Self-Preference Bias in Rubric-Based Evaluation of Large Language Models

José Pombal^{1,2,3}, Ricardo Rei¹ & André F. T. Martins^{2,3,4,5}

¹Sword Health, ²Instituto de Telecomunicações

³Instituto Superior Técnico, Universidade de Lisboa, ⁴TransPerfect, ⁵ELLIS Unit Lisbon
j.pombal@swordhealth.com

Abstract

LLM-as-a-judge has become the *de facto* approach for evaluating LLM outputs. However, judges are known to exhibit *self-preference bias* (SPB): they tend to favor outputs produced by themselves or by models from their own family. This skews evaluations and, thus, hinders model development, especially in settings of recursive self-improvement. We present the first study of SPB in rubric-based evaluation, an increasingly popular benchmarking paradigm where judges issue binary verdicts on individual evaluation criteria, instead of assigning holistic scores or rankings. Using IFEval and LiveCodeBench, benchmarks with programmatically verifiable rubrics, we show that SPB persists even when evaluation criteria are entirely objective: among rubrics where generators fail, judges can be more than 50% more likely to incorrectly mark them as satisfied when the output is their own. We also find that, similarly to other evaluation paradigms, ensembling multiple judges helps mitigate SPB, but without fully eliminating it. On HealthBench, a medical chat benchmark with subjective rubrics, we observe that SPB skews model scores by up to 10 points, a potentially decisive margin when ranking frontier models. We analyze the factors that drive SPB in this setting, finding that negative rubrics and subjective topics like communication and emergency referrals are particularly susceptible.

1 Introduction

Automated evaluation via LLM-as-a-judge has become the standard practice for assessing model outputs (Zheng et al., 2023). In this paradigm, an LLM is typically prompted to either compare two outputs and select the best one (pairwise comparison, PWC), or to rate an output in isolation according to a predefined numeric scale (direct assessment, DA). However, among other issues, self-preference bias (SPB) can occur under both approaches, whereby judges show a systematic preference for, or attribute disproportionately higher scores to outputs produced by themselves, by models from their own family, or by models trained on data they generated or curated (Verga et al., 2024; Dietz et al., 2025; Li et al., 2025; Spiliopoulou et al., 2025; Zhao et al., 2024; Pugachev et al., 2025; Wataoka et al., 2024; Koo et al., 2023). SPB may skew evaluations and misguide practitioners into believing certain systems are better than others. In self-improvement pipelines, SPB can lead to a feedback loop where models are optimized to perform well according to their own preferences, rather than truly improving on the task at hand (Xu et al., 2024; Panickssery et al., 2024).

More recently, rubric-based evaluation (RB) has emerged as an increasingly popular approach for benchmarking (Arora et al., 2025; Starace et al., 2025; Lin et al., 2024; Sirdeshmukh et al., 2025) and reward modeling in reinforcement learning (Huang et al., 2025; Gunjal et al., 2025; Dineen et al., 2025). In this setting, the judge is presented with one or more binary rubrics an output should satisfy (see Figure 1). Rubric-based evaluation allows for more granular and interpretable evaluations than PWC and DA, and has been shown to correlate well with human judgments (Arora et al., 2025; Starace et al., 2025; Lin et al., 2024; Sirdeshmukh et al., 2025). Furthermore, by discarding the need for direct comparisons

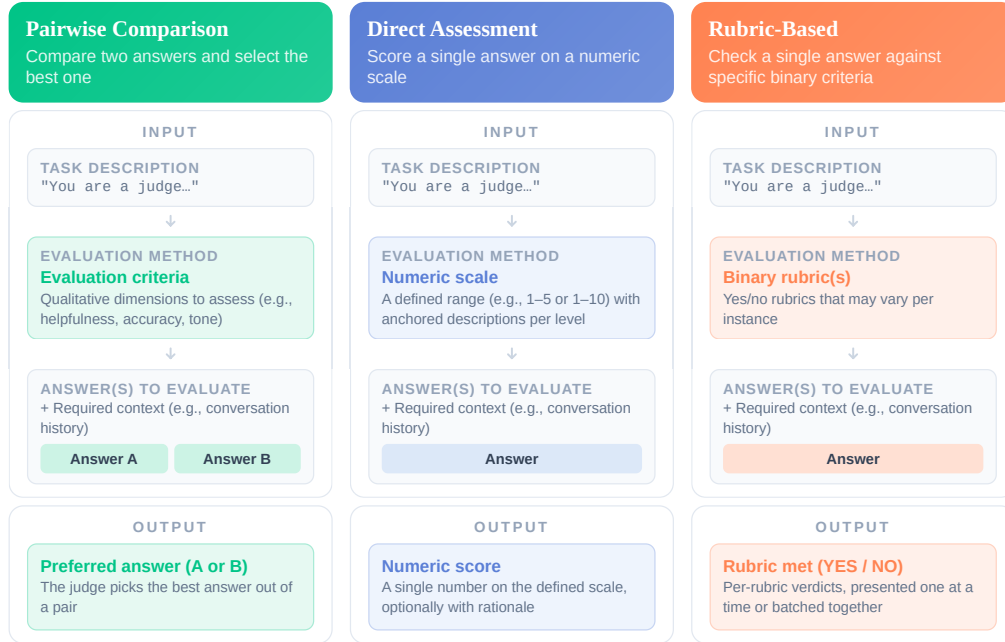


Figure 1: Illustration of three LLM-as-a-Judge paradigms: pairwise comparison (PWC), direct assessment (DA) and rubric-based evaluation (RB).

between outputs and for numeric scores, it eliminates the response ordering and score calibration issues found in PWC and DA, respectively. For the same reasons, RB could also be less susceptible to SPB, though this has not been systematically studied.

In this work, we first show that judges do, in fact, exhibit systematic preference for their own outputs—or those of their model family—in rubric-based evaluation, even when rubrics are entirely objective (*i.e.*, programmatically verifiable), and when controlling for instance difficulty and judge quality. For outputs that fail to satisfy a given rubric, models can be more than 50% more likely to incorrectly mark it as satisfied when the output was generated by themselves or their model family. Furthermore, the extent of self-preference bias in rubric-based evaluation is similar to that of direct assessment but significantly less severe than that of pairwise comparison, and, while the popular approach of ensembling multiple judges to mitigate SPB also proves effective, it does not fully eliminate the bias. We reach similar conclusions analyzing HealthBench (Arora et al., 2025), a realistic benchmark for medical chat assistant evaluation with subjective human-written rubrics. GPT-5, for example, tends to overestimate its performance by about 4 points relative to an ensemble of judges, a difference that may prove substantial when ranking frontier models and making claims about their capabilities.

We analyze the factors that contribute to SPB in HealthBench, uncovering several actionable patterns: (1) negative rubrics, which penalize generators if fulfilled, are more vulnerable to SPB than positive ones, (2) rubrics targeting communication and emergency referrals, among other themes, elicit higher SPB than those assessing accuracy, (3) inter-judge agreement can serve as a practical signal for flagging high-SPB rubrics. Together, these findings offer concrete guidance for designing rubrics that are more resilient to self-preference bias and point to promising directions for its mitigation.¹

¹We make code to reproduce our experiments and analysis available on [Github](#).

2 Experimental setup

2.1 Measuring self-preference bias

Self-preference bias (SPB) is the tendency of judges to unjustly and systematically favor their own outputs. To define a metric for SPB, we first need to define what *unjustly favoring* means, since judges may have legitimate reasons to prefer their own outputs, or a general tendency to over- or under-estimate any generator.

Generically, we can characterize a judge’s disposition toward a particular generator through its *overestimation rate*: the proportion of cases in which the judge rules more favorably for a generator than the ground truth warrants. Concretely, let $\mathcal{O}(\mathcal{J}, \mathcal{G})$ denote the non-negative overestimation rate of judge $\mathcal{J}$ toward generator $\mathcal{G}$, computed over a benchmark $\mathcal{D}$ (we define $\mathcal{O}$ precisely below for each evaluation paradigm in Figure 1). Self-preference then corresponds to the overestimation rate when the generator is the judge itself, $\mathcal{O}(\mathcal{J}, \mathcal{J})$. However, a high $\mathcal{O}(\mathcal{J}, \mathcal{J})$ alone does not constitute bias: the judge may simply be a lenient evaluator across the board. The concern arises when the judge overestimates its own outputs *disproportionately* relative to those of unrelated models. We capture this through the *Harmful Self-Preference Propensity Ratio* (HSPP Ratio, an extension of HSPP (Chen et al., 2025)):

$$\text{HSPP-R}_{\text{self}}(\mathcal{J}) = \frac{\mathcal{O}(\mathcal{J}, \mathcal{J})}{\frac{1}{|\mathcal{S}_{\mathcal{J}}|} \sum_{\mathcal{G} \in \mathcal{S}_{\mathcal{J}}} \mathcal{O}(\mathcal{J}, \mathcal{G})}, \quad (1)$$

where $\mathcal{S}_{\mathcal{J}}$ is the set of generators that are not $\mathcal{J}$ and do not belong to the same model family as $\mathcal{J}$. We exclude same-family generators from $\mathcal{S}_{\mathcal{J}}$ to prevent family-level biases from skewing self-preference. An HSPP-R of 1 indicates no self-preference; values above 1 indicate the judge overestimates its own outputs more than those of unrelated models.

Since models within the same family are often similar, SPB may extend beyond the judge itself to its relatives.² We therefore also define a *family* variant:

$$\text{HSPP-R}_{\text{fam}}(\mathcal{J}) = \frac{\frac{1}{|\mathcal{F}_{\mathcal{J}}|} \sum_{\mathcal{G} \in \mathcal{F}_{\mathcal{J}}} \mathcal{O}(\mathcal{J}, \mathcal{G})}{\frac{1}{|\mathcal{S}_{\mathcal{J}}|} \sum_{\mathcal{G} \in \mathcal{S}_{\mathcal{J}}} \mathcal{O}(\mathcal{J}, \mathcal{G})}, \quad (2)$$

where $\mathcal{F}_{\mathcal{J}}$ is the set of generators from the same family as $\mathcal{J}$ (excluding $\mathcal{J}$ to isolate family-only effects), and the denominator is the same as in Equation 1. The set of all generators is the disjoint union $\{\mathcal{J}\} \cup \mathcal{F}_{\mathcal{J}} \cup \mathcal{S}_{\mathcal{J}}$.

Instance-level overestimation. For evaluation methods that produce a scalar score per instance, we compare the judge’s instance-level orderings against the ground-truth orderings. This score may be derived from rubric satisfaction fractions (when the judge evaluates each rubric individually (single-rubric, SR), or all at once (all-rubrics, AR)), from a direct assessment scale (DA), or from pairwise comparison outcomes (PWC). For each instance $x \in \mathcal{D}$, let $s_{\mathcal{J}}(\mathcal{G}, x)$ be the judge’s score for generator $\mathcal{G}$ on instance x , and let $s^*(\mathcal{G}, x)$ be the corresponding ground-truth score.³ Let $\mathcal{G}' \neq \mathcal{G}$ be an opponent generator, and let the judge outcome be $w_{\mathcal{J}}(\mathcal{G}, \mathcal{G}', x) = \text{sgn}(s_{\mathcal{J}}(\mathcal{G}, x) - s_{\mathcal{J}}(\mathcal{G}', x))$ and the reference outcome $w^*(\mathcal{G}, \mathcal{G}', x) = \text{sgn}(s^*(\mathcal{G}, x) - s^*(\mathcal{G}', x))$, where $\text{sgn}(a)$ is $+1, 0$, or -1 according to whether a is positive ($\mathcal{G}$ wins), zero (tie), or negative ($\mathcal{G}$ loses). We define overestimation as the false-positive rate of the judge on instances where $\mathcal{G}$ should lose (i.e., $w^*(\mathcal{G}, \mathcal{G}', x) = -1$),

²We define “relatives” as models belonging to the same family (e.g., Gemma 3 27B and 12B). However, most models likely share some degree of *relatedness*, due to distillation or similar training data. How to best define and measure *relatedness* remains an open question.

³For SR and AR, s is the fraction of satisfied rubrics; for DA, we prompt the judge for the fraction of satisfied rubrics directly; for PWC, the resolved outcome is mapped to $\{+1, 0, -1\}$ for win, tie, loss.

measuring how often the judge nonetheless favors $\mathcal{G}$ (i.e., $w_{\mathcal{J}}(\mathcal{G}, \mathcal{G}', x) > w^*(\mathcal{G}, \mathcal{G}', x)$):

$$\mathcal{O}_{\text{inst}}(\mathcal{J}, \mathcal{G}) = \frac{\sum_{\mathcal{G}' \neq \mathcal{G}} \sum_{x \in \mathcal{D}} \mathbb{1}\{w_{\mathcal{J}}(\mathcal{G}, \mathcal{G}', x) > w^*(\mathcal{G}, \mathcal{G}', x), w^*(\mathcal{G}, \mathcal{G}', x) = -1\}}{\sum_{\mathcal{G}' \neq \mathcal{G}} \sum_{x \in \mathcal{D}} \mathbb{1}\{w^*(\mathcal{G}, \mathcal{G}', x) = -1\}}, \quad (3)$$

Rubric-level overestimation. For methods that produce individual rubric judgments (SR and AR), we can measure overestimation at a finer granularity. Let $b_{\mathcal{J}}(\mathcal{G}, x, k) \in \{-1, 1\}$ be the judge’s verdict on rubric k for generator $\mathcal{G}$ on instance x , and $b^*(\mathcal{G}, x, k)$ the ground-truth verdict. A rubric-level overestimation occurs when the judge marks a rubric as satisfied ($b_{\mathcal{J}} = 1$) while the ground truth marks it as unsatisfied ($b^* = -1$):

$$\mathcal{O}_{\text{rub}}(\mathcal{J}, \mathcal{G}) = \frac{\sum_{x \in \mathcal{D}} \sum_k \mathbb{1}\{b_{\mathcal{J}}(\mathcal{G}, x, k) = 1 \wedge b^*(\mathcal{G}, x, k) = -1\}}{\sum_{x \in \mathcal{D}} \sum_k \mathbb{1}\{b^*(\mathcal{G}, x, k) = -1\}}. \quad (4)$$

This is the false positive rate of the judge restricted to the outputs of a single generator and can be interpreted as: among rubrics that a generator objectively fails, how often does the judge incorrectly mark them as passed? Substituting either $\mathcal{O}_{\text{inst}}$ or $\mathcal{O}_{\text{rub}}$ into Equations 1–2 yields the instance-level or rubric-level HSPP Ratio, respectively.

2.2 Measuring performance

Accuracy measures the rate of agreement between the judge’s verdicts and the reference across a benchmark $\mathcal{D}$. As with SPB, this can be instantiated at different granularities depending on the evaluation paradigm.

Instance-level accuracy. For all four paradigms (SR, AR, DA, PWC), we measure accuracy through *pairwise concordance*: for each judge, instance, and pair of generators $(\mathcal{G}, \mathcal{G}')$, we check whether the judge’s ordering agrees with the reference. Using the same notation as in Equation 3, a comparison is concordant if $w_{\mathcal{J}}(\mathcal{G}, \mathcal{G}', x) = w^*(\mathcal{G}, \mathcal{G}', x)$, including ties (i.e., both signs are zero). The *Mean Instance Pairwise Accuracy* (MIPA) for judge $\mathcal{J}$ is:

$$\text{MIPA}(\mathcal{J}) = \frac{1}{\binom{|\mathcal{G}|}{2} |\mathcal{D}|} \sum_{\mathcal{G} < \mathcal{G}'} \sum_{x \in \mathcal{D}} \mathbb{1}\{w_{\mathcal{J}}(\mathcal{G}, \mathcal{G}', x) = w^*(\mathcal{G}, \mathcal{G}', x)\}, \quad (5)$$

where the outer sum runs over all unordered pairs of generators. For PWC, we run each comparison twice, swapping the presentation order of the two outputs to mitigate position bias (Zheng et al., 2023). If one run produces a winner and the other is a tie, the non-tie result stands. If both runs are ties or the two runs disagree on the winner, the result is a tie.

Rubric-level accuracy. For SR and AR, the *Mean Rubric Accuracy* (MRA) for judge $\mathcal{J}$ is simply the fraction of individual rubric verdicts that match the reference:

$$\text{MRA}(\mathcal{J}) = \frac{\sum_{\mathcal{G}} \sum_{x \in \mathcal{D}} \sum_k \mathbb{1}\{b_{\mathcal{J}}(\mathcal{G}, x, k) = b^*(\mathcal{G}, x, k)\}}{\sum_{\mathcal{G}} \sum_{x \in \mathcal{D}} \sum_k 1}. \quad (6)$$

2.3 Statistical significance

We report 95% confidence intervals for all metrics, computed via percentile bootstrap with 1000 resamples and shown in gray next to each point estimate. The resampling unit matches the granularity of the metric: for instance-level metrics, we resample benchmark instances; for rubric-level metrics, we pool the eligible rubric verdicts of all participating generators and resample from this global pool, recomputing the full metric, numerator and denominator alike, on each resample. When comparing paired quantities (e.g., a committee member against the committee on the member’s own outputs, Appendix F), we share resamples across the two sides so that the interval on their difference is valid.

2.4 Models

We evaluate a broad set of frontier open- and closed-weights models spanning different families and sizes as judges and generators: Gemma 3 27B, 12B, and 4B (Kamath et al., 2025), Llama 4 Maverick and Scout (Meta, 2025), Qwen 3 235B, 30B, and 4B (Yang et al., 2025), GPT-5 and GPT-oss-120B (Agarwal et al., 2025), and Claude 4.5 Sonnet and Haiku.⁴

2.5 Datasets

The metrics in Section 2.1 require a ground-truth reference, which is typically unavailable; this is precisely why LLM-as-a-judge is used in the first place. Thus, inspired by Chen et al. (2025), we first conduct our analysis on two benchmarks: 1) IFEval (Zhou et al., 2023), a benchmark of 541 prompts for evaluating instruction-following, where each prompt is paired with one or more programmatically verifiable instructions (e.g., “no commas”); and 2) LiveCodeBench (Jain et al., 2025), a code-generation benchmark.⁵ We convert each instruction and each private unit test into binary rubrics, respectively. This allows us to compute overestimation metrics precisely and to isolate the effect of SPB from confounders like rubric subjectivity (Section 3).

To analyze SPB in a subjective setting, we turn to HealthBench (Arora et al., 2025) in Section 4, a benchmark for medical chat assistant evaluation containing 5,000 instances and 48,562 unique binary rubrics. Rubrics range from accuracy and instruction-following to highly subjective areas like communication tone, and are each associated with a weight toward the final score that the judge does not have access to. Some rubrics are negative, meaning that the generator is penalized for satisfying them. Since ground-truth verdicts are unavailable, we use as reference a majority vote among the largest judge of each family (Gemma 3 27B, Llama 4 Maverick, Qwen 3 235B, GPT-oss-120B, and Claude Sonnet 4.5). All 5 models score higher than 0.6 on HealthBench’s own meta-evaluation set (Appendix C), which is within the range of solid judge performance reported by the authors. While imperfect, this is nonetheless informative: ensembles of judges have been shown to be more accurate and robust to SPB than individual judges (Verga et al., 2024), which we observe on IFEval (§3).⁶ Appendix A contains details on the prompts used for all evaluation modes and benchmarks.

3 Self-preference bias with objective rubrics

Judges favor themselves and models from their family even with objective rubrics. The LiveCodeBench part of Table 1 shows that most judges have a rubric-level HSPP ratio significantly greater than 1. In the worst case (GPT-5), the judge can be 20 times more likely to overestimate its own outputs than those of unrelated models. On IFEval, all but one judge exhibits a mean HSPP ratio above 1, though statistical significance only holds for two systems. Table 5 shows similar results at the family level. This finding suggests that perfect evaluation criteria alone are not sufficient to prevent SPB at individual or family levels. Elimination of SPB may instead require more intricate strategies (e.g., model training, activation steering (Roytburg et al., 2025)), rather than just careful evaluation design.

Rubric-based evaluation is more robust than pairwise comparison, but not necessarily direct assessment. Figure 2 shows that, while SPB is present across all paradigms, it is significantly more severe in PWC than in RB and DA. This could be because PWC directly pits the judge’s output against that of an opponent, making it easier for the judge to identify and favor its own output. Indeed, Panickssery et al. (2024) have shown that SPB is related

⁴We use proprietary models with default reasoning effort, and we do not include them in the AR, DA, and PWC experiments for cost purposes. In Appendix G, we find that reasoning effort does not reduce self-preference on IFEval.

⁵We use the data split after 2025-01-04 to limit data contamination, and consider only tests that every model managed to judge, yielding 177 problems and 5,685 distinct tests/rubrics.

⁶Using an ensemble as a reference actually leads to an underestimation of SPB, considering that all model families are represented in the reference (see Appendix J).

Judge	IFEval		LiveCodeBench		HealthBench	
	MRA	HSPP-Rub. (Self)	MRA	HSPP-Rub. (Self)	MRA	HSPP-Rub. (Self)
•Gemma-4B	0.89 [0.89, 0.90]	1.03 [1.00, 1.06]	0.46 [0.45, 0.46]	1.54 [1.38, 1.73]	0.71 [0.70, 0.71]	1.16 [1.14, 1.17]
•Gemma-12B	0.89 [0.88, 0.89]	1.03 [0.95, 1.11]	0.74 [0.74, 0.75]	0.97 [0.93, 1.01]	0.77 [0.77, 0.77]	1.03 [1.01, 1.05]
•Gemma-27B	0.89 [0.89, 0.90]	1.08 [1.02, 1.14]	0.77 [0.77, 0.78]	1.03 [0.97, 1.08]	0.83 [0.83, 0.83]	1.10 [1.08, 1.12]
•Llama-Scout	0.84 [0.83, 0.85]	1.05 [0.91, 1.20]	0.77 [0.77, 0.77]	1.17 [1.14, 1.20]	0.83 [0.83, 0.83]	0.80 [0.79, 0.82]
•Llama-Mav	0.88 [0.87, 0.88]	1.12 [0.98, 1.27]	0.79 [0.78, 0.79]	1.34 [1.30, 1.38]	0.90 [0.90, 0.90]	0.71 [0.69, 0.73]
•Qwen-4B	0.85 [0.84, 0.86]	1.06 [0.92, 1.20]	0.67 [0.66, 0.67]	2.21 [2.03, 2.42]	0.77 [0.77, 0.77]	0.96 [0.95, 0.98]
•Qwen-30B	0.88 [0.88, 0.89]	1.05 [0.93, 1.15]	0.79 [0.78, 0.79]	1.08 [0.99, 1.18]	0.81 [0.81, 0.81]	1.06 [1.04, 1.08]
•Qwen-235B	0.89 [0.88, 0.89]	1.30 [1.11, 1.48]	0.65 [0.65, 0.66]	0.83 [0.62, 1.07]	0.94 [0.94, 0.94]	1.12 [1.07, 1.18]
•GPT-120B	0.91 [0.90, 0.91]	1.30 [0.81, 1.85]	0.91 [0.91, 0.92]	5.97 [4.91, 7.18]	0.87 [0.87, 0.87]	1.00 [0.94, 1.06]
•GPT-5	0.91 [0.91, 0.92]	1.47 [0.48, 2.53]	0.97 [0.97, 0.97]	20.15 [16.57, 24.58]	0.86 [0.86, 0.86]	1.54 [1.44, 1.64]
•Claude-Haiku	0.81 [0.80, 0.82]	1.15 [0.88, 1.44]	0.92 [0.92, 0.93]	1.71 [1.57, 1.85]	0.84 [0.84, 0.84]	0.90 [0.86, 0.93]
•Claude-Sonnet	0.84 [0.83, 0.85]	0.98 [0.64, 1.32]	0.95 [0.95, 0.95]	1.78 [1.58, 1.99]	0.88 [0.88, 0.88]	0.93 [0.86, 1.01]

Table 1: Rubric-level metrics per judge across IFEval, LiveCodeBench, and HealthBench (SR), with 95% confidence intervals (rubric-level bootstrap, 1000 resamples) in grey. MRA = Mean Rubric Accuracy; HSPP-Rub. (Self) = rubric-level HSPP ratio for self-generated outputs. Family-level ratios are in Appendix D.

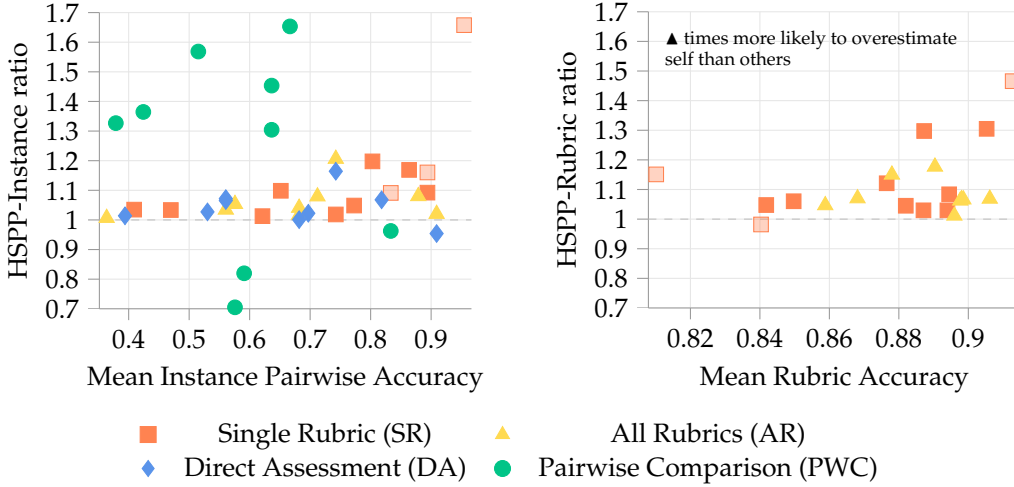


Figure 2: Performance vs. self-preference bias on IFEval across judge paradigms. SR shows also shows GPT-5 and Claude models (light orange squares), which we did not run in other methods for cost purposes. *Left*: Instance-level (MIPA vs. HSPP-Instance ratio). *Right*: Rubric-level (MRA vs. HSPP-Rubric ratio).

to the judge’s ability to identify its own outputs, which we also find in Appendix I. At the rubric level, presenting one rubric at a time elicits slightly higher SPB than presenting all at once, possibly because the narrow scope makes it easier for the judge to rationalize away failures on its own outputs.

Ensembling judgments helps mitigate and predict self-preference. In Figure 3, we show instance- and rubric-level accuracy and SPB metrics on IFEval for the largest judge of each family, as well as for a majority-vote ensemble of all five judges. A judge’s committee-aggregated performance and SPB are computed in the same way as for individual judges, but using the ensemble’s majority vote as the prediction instead of the individual verdict.

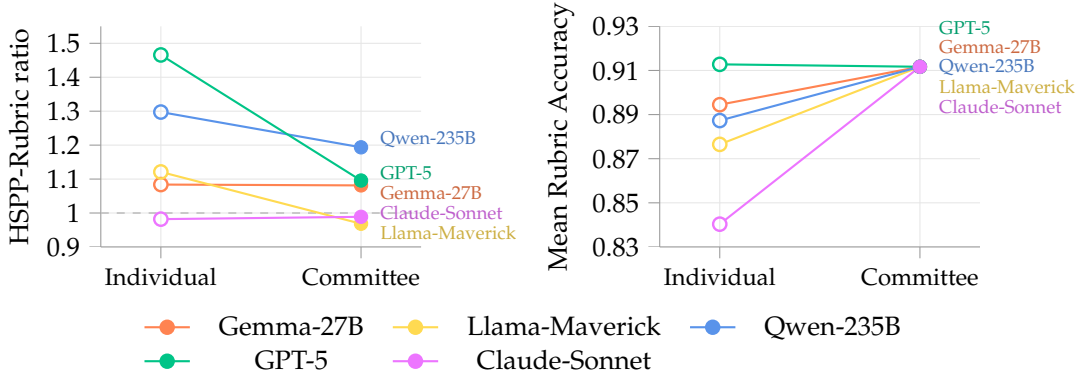


Figure 3: Impact of a 5-family committee on rubric-level self-preference bias (left) and accuracy (right) on IFEval (SR). Lines connect each committee member’s individual performance to its committee-aggregated performance.



Figure 4: HSPP-Instance (left) and HSPP-Rubric (right) when filtering rubrics by inter-judge pairwise agreement threshold on IFEval (SR, 12 judges, mean over all judges).

Across the board, ensembling leads to the same or higher accuracy and lower SPB, though it does not fully eliminate SPB. This is in line with the findings of Verga et al. (2024) and suggests that ensembling is a practical strategy for mitigating SPB, though not sufficient on its own. In Appendix F, we show that the main drivers of ensemble-based mitigation are the inclusion of high-quality judges, and, to a lesser extent, family diversity. Furthermore, we find that removing rubrics with low inter-judge agreement can be an effective way to reduce SPB, as shown in Figure 4, indicating that agreement can be a useful signal for flagging problematic rubrics.

Self-preference resists generator capability confounds. One possible explanation for the high HSPP values of models like GPT-5 on LiveCodeBench is that they are simply better at the task, so their mistakes are harder to judge and, therefore, more likely to be overestimated. To control for this, we stratify instances and their respective completion by mean rubric accuracy across all 12 judges, which serves as a proxy for their “hardness”. Concretely, instances where judges correctly classify more rubrics are considered easier, while those that judges incorrectly classify more rubrics wrong are considered harder. We expect better generators to have a higher fraction of failed unit tests in the harder bins, and worse generators to have more failed tests in the easier bins; this is indeed the case, as we show in Appendix Table 16. Thus, in Table 2, we recompute the rubric-level HSPP ratio for the 4 best-performing models on LiveCodeBench. If self-preference were an artifact of hard-to-judge generations, the ratio should collapse to 1 within each hardness bin. Instead, we observe that, with the exception of Qwen 235B, models maintain a high HSPP ratio across bins, indicating that self-preference persists beyond generator capability.

Judge	Original	Mean-MRA bin (proxy for evaluation “hardness”)			
		0.5–0.6	0.6–0.7	0.7–0.8	0.8–0.9
•GPT-5	20.15 [16.57, 24.58]	13.74 [9.94, 19.29]	14.66 [8.94, 21.48]	7.36 [1.50, 15.55]	0.00 [0.00, 0.00]
•GPT-120B	5.97 [4.91, 7.18]	1.13 [0.59, 1.84]	1.98 [0.83, 3.24]	3.98 [1.48, 7.33]	7.35 [1.34, 21.10]
•Claude-Sonnet	1.78 [1.58, 1.99]	1.86 [1.56, 2.14]	1.33 [1.11, 1.60]	2.06 [1.45, 2.89]	2.76 [0.89, 5.22]
•Qwen-235B	0.83 [0.62, 1.07]	0.78 [0.40, 1.23]	0.93 [0.58, 1.38]	0.25 [0.06, 0.61]	2.28 [0.79, 4.49]

Table 2: LiveCodeBench rubric-level HSPP (Self) stratified by generation hardness: ratios are recomputed using only (generator, instance) cells whose mean rubric accuracy across all 12 judges falls in the bin. “Original” is the unstratified ratio from Table 1. Null values occur when there are no instances in the bin for that generator.

Generator → Judge ↓	Gemma-4B	Gemma-12B	Gemma-27B	Llama-Scout	Llama-Mav	Qwen-4B	Qwen-30B	Qwen-235B	GPT-120B	GPT-5	Claude-Haiku	Claude-Sonnet
Gemma-4B	10.90	8.46	5.81	1.55	0.48	0.82	-2.07	-5.33	3.85	-12.53	-4.99	-6.94
Gemma-12B	3.05	-2.07	-2.37	5.92	5.42	1.86	-0.58	-2.77	-5.72	-8.14	3.60	1.79
Gemma-27B	4.40	-0.62	-1.08	4.43	4.11	1.72	-0.56	-2.28	-5.20	-7.52	2.31	0.27
Llama-Scout	3.23	-1.06	-0.65	2.62	1.70	2.51	0.71	-0.71	-3.56	-5.22	1.10	-0.69
Llama-Mav	3.27	0.66	0.01	0.41	-0.19	1.03	-0.06	-1.43	-0.91	-3.26	0.57	-0.11
Qwen-4B	4.06	-1.93	-2.49	6.93	7.12	0.88	-1.64	-2.83	-4.94	-7.44	2.36	-0.07
Qwen-30B	4.86	-0.75	-1.33	3.97	3.37	2.44	-0.49	-2.27	-4.36	-6.37	1.58	-0.65
Qwen-235B	-0.45	-0.28	-0.17	0.43	0.67	0.19	0.35	0.22	-0.05	-0.35	-0.29	-0.26
GPT-120B	-0.77	-1.59	-1.75	2.77	3.22	0.38	-0.16	-0.32	-1.05	-1.96	0.93	0.31
GPT-5	-2.11	-2.49	-2.63	0.72	2.49	-2.29	-0.75	1.05	-1.52	4.63	0.88	2.01
Claude-Haiku	-0.17	-0.64	-0.83	-1.70	-1.18	-0.13	0.82	0.58	0.85	0.97	0.96	0.47
Claude-Sonnet	-1.47	0.06	-0.08	0.02	1.39	-3.87	-0.83	1.09	-0.32	1.87	0.61	1.52

Figure 5: Centered score delta matrix for HealthBench (SR, weighted scoring), scaled by $\times 100$. Each cell shows judge j ’s deviation for generator g , centered by j ’s average bias. Orange = relative overestimation; blue = relative underestimation. Bordered cells indicate self-evaluation, while dotted areas represent within-family evaluation. See confidence intervals in Table 28.

4 Self-preference bias with subjective rubrics

Self-preference bias can significantly skew model scores on HealthBench. The HealthBench side of Table 1 further suggests that SPB persists with subjective rubrics. Figure 5 shows the centered score delta matrix for assessing the practical impact of SPB on evaluating models. For each judge-generator pair, we compute the difference between the judge’s system-level score and the reference score, then subtract the judge’s mean difference across all generators to control for overall leniency or strictness. Each cell thus reflects how much a judge over- or under-estimates a specific generator relative to its own baseline. Family-level effects are visible: Gemma judges tend to overestimate other Gemma models, particularly Gemma-4B, whose self-overestimation reaches nearly 11 points. The Llama family, by contrast, shows no strong self- or family-preference on HealthBench, consistent with its below-1 HSPP ratios in Table 1. Perhaps most importantly, judges exhibit a clear diagonal pattern: GPT-5, for instance, assigns itself a relative bonus of approximately 4 points (on a 100-point scale), a margin that could prove decisive when ranking frontier models.

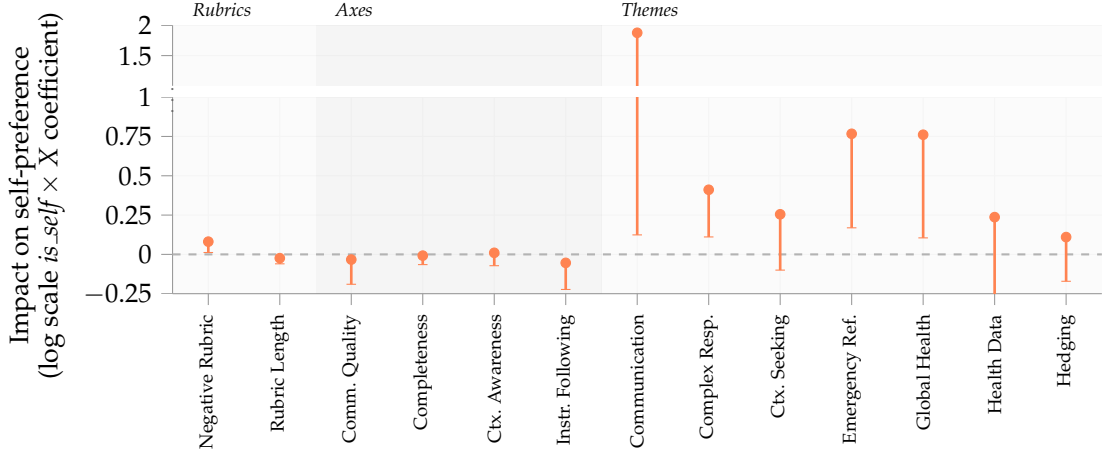


Figure 6: Interaction coefficients $is_self \times X$ (posterior mean) of a mixed-effects logistic regression of rubric-level overestimation on HealthBench, with random intercepts on judge and instance (Appendix K). Positive values mean property X amplifies overestimation *more when the judge evaluates its own output*, i.e., it reinforces self-preference. Left to right: rubric properties (negative polarity, rubric length); rubric axes (reference = Accuracy); themes (reference = untaged). For readability, only the lower half of each 95% interval is drawn and the y -axis is compressed above 1; full intervals are in Table 24.

Negative rubrics and certain rubric topics reinforce overestimation. To disentangle which rubric properties drive self-preference, we fit a mixed-effects logistic regression on HealthBench: for every rubric, we model the probability that the judge overestimates it. Fixed effects comprise whether the output under evaluation is the judge’s own (is_self) or from its family, the rubric’s polarity (positive or negative), its (standardized log) length, and one indicator per axis and theme. Furthermore, we include interaction terms between every fixed effect and is_self , which capture their impact on self-preference. We add random intercepts on judge and instance absorb each judge’s baseline leniency and each instance’s difficulty. Figure 6 shows the posterior mean and 95% interval of each interaction term. Negative rubrics consistently increase the odds of overestimation, though by a smaller margin than certain themes, like communication, global health, and emergency referalls. The instruction-following axis, which represents more verifiable rubrics, elicit comparatively low bias. All in all, these findings indicate that overestimation varies widely across rubric types, and reinforce the importance of rubric design across applications.

5 Related work

5.1 Rubric-based evaluation

LLM-as-a-judge (Zheng et al., 2023) has become the dominant paradigm for automatic evaluation of LLM outputs. Most work in this space has focused on pairwise comparison (PWC), where a judge selects the better of two outputs, and direct assessment (DA), where a judge rates a single output on a numeric scale. More recently, rubric-based evaluation (RB) has gained traction as an alternative that replaces holistic judgments with binary, per-criterion verdicts. RB has been adopted in prominent benchmarks spanning instruction-following (Lin et al., 2024; Sirdeshmukh et al., 2025), research replication (Starace et al., 2025), and medical chat evaluation (Arora et al., 2025). Rubric-based approaches have also been applied to reward modelling in reinforcement learning, where per-rubric verdicts serve as fine-grained reward signals (Huang et al., 2025; Gunjal et al., 2025; Dineen et al., 2025). A key advantage of RB is that each instance can have its own set of success criteria, enabling more granular and interpretable evaluations than PWC or DA. Furthermore, by reducing

the judge’s task to a series of binary decisions, RB eliminates the need for direct output comparisons and numeric score calibration, addressing core failure modes of PWC and DA, respectively (Zheng et al., 2023). Several works have shown that RB correlates well with human judgments (Arora et al., 2025; Starace et al., 2025; Lin et al., 2024; Sirdeshmukh et al., 2025), further supporting its adoption. However, whether RB inherits the self-preference bias observed in PWC and DA has not been systematically studied, a gap that our work aims to fill.

5.2 Self-preference bias

Self-preference bias (SPB) in LLM evaluators has attracted growing attention (Koo et al., 2023; Wataoka et al., 2024; Chen et al., 2025; Li et al., 2025; Spiliopoulou et al., 2025; Pugachev et al., 2025; Zhao et al., 2024; Dietz et al., 2025; Verga et al., 2024; Roytburg et al., 2025). For instance, Koo et al. (2023) identify SPB as one of several cognitive biases exhibited by LLM judges. Panickssery et al. (2024) study SPB in pairwise comparison and find that it is correlated with the judge’s ability identify its own outputs. Chen et al. (2025) propose the Harmful Self-Preference Propensity (HSPP) metric, which restricts the analysis to cases where the judge’s own output is objectively worse, and find that even in these cases judges tend to favor themselves. Li et al. (2025) show that judges can even favor outputs of models trained on synthetic data generated by themselves. On the mitigation side, Verga et al. (2024) propose replacing a single judge with a jury several models. Roytburg et al. (2025) use activation steering to directly suppress self-preference at inference time. All the aforementioned works focus on PWC and DA, which are being progressively replaced by rubric-based evaluation methods. We show that SPB persists in RB—even with fully objective rubrics—and that ensembling multiple judges, while effective, does not fully eliminate it, suggesting that more intricate mitigation strategies may be necessary. We also find that SPB correlates with self-recognition ability and output length, but not as much with perplexity (see Appendix I).

6 Conclusion

We presented the first systematic study of self-preference bias in rubric-based evaluation, showing that it persists even with fully objective rubrics and beyond generator capability confounds, and can meaningfully skew benchmark scores in subjective settings. Rubric-based evaluation is significantly more robust than pairwise comparison but not immune to SPB, and ensembling judges helps but does not fully eliminate it. We identified several rubric-level factors that drive SPB, offering concrete guidance for more resilient evaluation design. Our findings suggest that SPB is not just an artifact of evaluation methodology, and that its full mitigation will likely require interventions beyond rubric construction, such as inference-time debiasing or training-time corrections. For future work, extending this analysis to reward modelling pipelines is a natural next step. We expect self-preference bias to accelerate reward hacking, making training plateau and degenerate faster. This would be particularly problematic for tasks that can only be solved and evaluated by a specific system, potentially hindering the development of specialized models.

Acknowledgments

This work is supported by the project DECOLLAGE (ERC-2022-CoG 101088763) and by FC-T/MECI through national funds and when applicable cofunded EU funds under UID/50008: Instituto de Telecomunicações. We also thank the COLM reviewers, whose feedback made the paper considerably stronger.

References

Sandhini Agarwal, Lama Ahmad, Jason Ai, Sam Altman, Andy Applebaum, Edwin Arbus, Rahul K Arora, Yu Bai, Bowen Baker, Haiming Bao, et al. gpt-oss-120b & gpt-oss-20b model card. *arXiv preprint arXiv:2508.10925*, 2025.

- Rahul K Arora, Jason Wei, Rebecca Soskin Hicks, Preston Bowman, Joaquin Quiñero-Candela, Foivos Tsimpourlas, Michael Sharman, Meghan Shah, Andrea Vallone, Alex Beutel, et al. Healthbench: Evaluating large language models towards improved human health. *arXiv preprint arXiv:2505.08775*, 2025.
- Wei-Lin Chen, Zhepei Wei, Xinyu Zhu, Shi Feng, and Yu Meng. Do llm evaluators prefer themselves for a reason? *arXiv preprint arXiv:2504.03846*, 2025.
- Laura Dietz, Oleg Zendel, Peter Bailey, Charles Clarke, Ellese Cotterill, Jeff Dalton, Faegheh Hasibi, Mark Sanderson, and Nick Craswell. Llm-evaluation tropes: Perspectives on the validity of llm-evaluations. *arXiv preprint arXiv:2504.19076*, 2025.
- Jacob Dineen, Aswin Rrv, Qin Liu, Zhikun Xu, Xiao Ye, Ming Shen, Zhaonan Li, Shijie Lu, Chitta Baral, Muhao Chen, and Ben Zhou. QA-LIGN: Aligning LLMs through constitutionally decomposed QA. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 20619–20642, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.1123. URL <https://aclanthology.org/2025.findings-emnlp.1123/>.
- Anisha Gunjal, Anthony Wang, Elaine Lau, Vaskar Nath, Bing Liu, and Sean Hendryx. Rubrics as rewards: Reinforcement learning beyond verifiable domains. *arXiv preprint arXiv:2507.17746*, 2025.
- Zenan Huang, Yihong Zhuang, Guoshan Lu, Zeyu Qin, Haokai Xu, Tianyu Zhao, Ru Peng, Jiaqi Hu, Zhanming Shen, Xiaomeng Hu, et al. Reinforcement learning with rubric anchors. *arXiv preprint arXiv:2508.12790*, 2025.
- Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. Livecodebench: Holistic and contamination free evaluation of large language models for code. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, Louis Rouillard, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 4, 2025.
- Ryan Koo, Minhwa Lee, Vipul Raheja, Jong Inn Park, Zae Myung Kim, and Dongyeop Kang. Benchmarking cognitive biases in large language models as evaluators. *arXiv preprint arXiv:2309.17012*, 2023.
- Dawei Li, Renliang Sun, Yue Huang, Ming Zhong, Bohan Jiang, Jiawei Han, Xiangliang Zhang, Wei Wang, and Huan Liu. Preference leakage: A contamination problem in llm-as-a-judge. *arXiv preprint arXiv:2502.01534*, 2025.
- Bill Yuchen Lin, Yuntian Deng, Khyathi Chandu, Faeze Brahman, Abhilasha Ravichander, Valentina Pyatkin, Nouha Dziri, Ronan Le Bras, and Yejin Choi. Wildbench: Benchmarking llms with challenging tasks from real users in the wild. *arXiv preprint arXiv:2406.04770*, 2024.
- Meta. The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/>, April 2025. Accessed: 2026-03-15.
- Arjun Panickssery, Samuel R Bowman, and Shi Feng. Llm evaluators recognize and favor their own generations. In *Advances in Neural Information Processing Systems*, volume 37, pp. 68772–68802, 2024.
- Alexander Pugachev, Alena Fenogenova, Vladislav Mikhailov, and Ekaterina Artemova. Repa: Russian error types annotation for evaluating text generation and judgment capabilities. *arXiv preprint arXiv:2503.13102*, 2025.

- Dani Roytburg, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Simon Fu, and Narmeen Oozeer. Breaking the mirror: Activation-based mitigation of self-preference in llm evaluators. *arXiv preprint arXiv:2509.03647*, 2025.
- Ved Sirdeshmukh, Kaustubh Deshpande, Johannes Mols, Lifeng Jin, Ed-Yeremai Cardona, Dean Lee, Jeremy Kritz, Willow Primack, Summer Yue, and Chen Xing. Multichallenge: A realistic multi-turn conversation evaluation benchmark challenging to frontier llms. *arXiv preprint arXiv:2501.17399*, 2025.
- Evangelia Spiliopoulou, Riccardo Fogliato, Hanna Burnsky, Tamer Soliman, Jie Ma, Graham Horwood, and Miguel Ballesteros. Play favorites: A statistical method to measure self-bias in llm-as-a-judge. *arXiv preprint arXiv:2508.06709*, 2025.
- Giulio Starace, Oliver Jaffe, Dane Sherburn, James Aung, Jun Shern Chan, Leon Maksin, Rachel Dias, Evan Mays, Benjamin Kinsella, Wyatt Thompson, et al. Paperbench: Evaluating ai’s ability to replicate ai research. *arXiv preprint arXiv:2504.01848*, 2025.
- Pat Verga, Sebastian Hofstatter, Sophia Althammer, Yixuan Su, Aleksandra Piktus, Arkady Arkhangorodsky, Minjie Xu, Naomi White, and Patrick Lewis. Replacing judges with juries: Evaluating llm generations with a panel of diverse models. *arXiv preprint arXiv:2404.18796*, 2024.
- Koki Wataoka, Tsubasa Takahashi, and Ryokan Ri. Self-preference bias in llm-as-a-judge. *arXiv preprint arXiv:2410.21819*, 2024.
- Xuyang Wu, Jinming Nian, Ting-Ruen Wei, Zhiqiang Tao, Hsin-Tai Wu, and Yi Fang. Does reasoning introduce bias? a study of social bias evaluation and mitigation in LLM reasoning. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 18534–18555, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.1006. URL <https://aclanthology.org/2025.findings-emnlp.1006/>.
- Wenda Xu, Guanglei Zhu, Xuandong Zhao, Liangming Pan, Lei Li, and William Yang Wang. Pride and prejudice: Llm amplifies self-bias in self-refinement. *arXiv preprint arXiv:2402.11436*, 2024.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Justin Zhao, Flor Miriam Plaza-del Arco, Benjamin Genchel, and Amanda Cercas Curry. Language model council: Democratically benchmarking foundation models on highly subjective tasks. *arXiv preprint arXiv:2406.08598*, 2024.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36:46595–46623, 2023.
- Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

A Prompts

In Figures 7 through 11 we show the prompts used for all evaluation modes and benchmarks. Crucially, regardless of the evaluation paradigm, judges receive, one way or another, the same amount of information regarding evaluation criteria.

IFEval — Single-Rubric (SR) Evaluation Prompt

Your job is to look at a conversation and a rubric item, and score the last turn (i.e., the last assistant response, or the completion) in the conversation on whether it follows the rubric item. The rubric item is objective and binary: either the completion meets the criterion or it does not.

Conversation
<<conversation>>

Rubric item
<<rubric_item>>

Instructions
Return a json object with "criteria_met" field.
- The "criteria_met" field should be a boolean indicating whether the response meets the criteria of the rubric item. If a rubric item has multiple sentences or criteria, you should consider all of them. If any of the criteria is not met, the answer should be false. Only return true if all of the criteria are met.

Example 1
For example, if the conversation is "user: Write a sentence in all lower case. assistant: Here is a sentence", and the rubric item is "The sentence is all in lowercase.", you should return a json like this you should return a json like this:

```
```json
{
 "criteria_met": false
}
```
```

Final instruction
Return just the json object in markdown format. Do not include any other text in the response.

Figure 7: Prompt template for single-rubric (SR) evaluation on IFEval.

IFEval — All-Rubrics (AR) Evaluation Prompt

Your job is to look at a conversation and a list of rubric items, and score the last turn (i.e., the last assistant response, or the completion) on whether it follows each rubric item. Each rubric item is objective and binary: either the completion meets the criterion or it does not.

Conversation
<<conversation>>

Rubric items
<<rubric_items>>

Instructions
Return a json object with a "results" field containing a list of objects, one per rubric item, in the same order as the rubric items above. Each object should have a "criteria_met" field (boolean).
- For each rubric item, "criteria_met" should be true only if ALL criteria within that rubric item are met; otherwise false.

Example
For example, if there are 3 rubric items and the response meets the first and third but not the second, you should return:

```
```json
{
 "results": [
 {"criteria_met": true},
 {"criteria_met": false},
 {"criteria_met": true}
]
}
```
```

Final instruction
Return just the json object in markdown format. Do not include any other text in the response.

Figure 8: Prompt template for all-rubrics (AR) evaluation on IFEval.

IFEval — Direct Assessment (DA) Evaluation Prompt

Your job is to look at a conversation and a set of rubric items, and provide an overall score for the last turn (i.e., the last assistant response, or the completion) based on how many of the rubric items are satisfied. Each rubric item is objective and binary: either the completion meets the criterion or it does not.

Conversation
<<conversation>>

Rubric items
<<rubric_items>>

Instructions
Consider all rubric items above. Determine how many of the rubric items are met by the response.
Return a json object with a "score" field.

- The "score" field should be a string representing the fraction of rubric items met, always in the format "X/Y" where X is the number of met rubric items and Y is the total number of rubric items.
- If all rubric items are met, return "Y/Y" (e.g., "3/3").
- If no rubric items are met, return "0/Y" (e.g., "0/3").
- If some but not all rubric items are met, return the appropriate fraction (e.g., "2/3").

Example
For example, if there are 3 rubric items and 2 are met, you should return:

```
```json
{
 "score": "2/3"
}
```
```

If all 3 are met:

```
```json
{
 "score": "3/3"
}
```
```

If none are met:

```
```json
{
 "score": "0/3"
}
```
```

Final instruction
Return just the json object in markdown format. Do not include any other text in the response.

Figure 9: Prompt template for direct assessment (DA) evaluation on IFEval.

IFEval — Pairwise Comparison (PWC) Evaluation Prompt

Your job is to compare two responses to a conversation, given a set of rubric items, and determine which response better satisfies the rubric items. Each rubric item is objective and binary: either the completion meets the criterion or it does not.

```
# Conversation
<<conversation>>

# Rubric items
<<rubric_items>>

<Response A>
<<response_a>>
</Response A>

<Response B>
<<response_b>>
</Response B>

# Instructions
Compare Response A and Response B based on how well each satisfies the rubric items above.
Return a json object with an "outcome" field.
- The "outcome" field should be "A is better" if Response A better satisfies the rubric items overall.
- The "outcome" field should be "B is better" if Response B better satisfies the rubric items overall.
- The "outcome" field should be "tie" if both responses satisfy the rubric items equally well (or equally poorly).

Consider ALL rubric items when making your judgment. A response that satisfies more rubric items should generally be preferred.
```

```
# Example 1
```json
{
 "outcome": "tie"
}
```

# Example 2
```json
{
 "outcome": "B is better"
}
```

# Example 3
```json
{
 "outcome": "A is better"
}
```

# Final instruction
Return just the json object in markdown format. Do not include any other text in the response.
```

Figure 10: Prompt template for pairwise comparison (PWC) evaluation on IFEval.

HealthBench — Single-Rubric (SR) Evaluation Prompt

Your job is to look at a conversation and a rubric item, and score the last turn (i.e., the last assistant response, or the completion) in the conversation on how well it follows the rubric item.

```
# Conversation
<<conversation>>

# Rubric item
<<rubric_item>>

# Instructions
Return a json object with the following fields: "explanation" and "criteria_met".
- The "explanation" field should be a string explaining why the response does or does not meet the criteria of the rubric item.
- The "criteria_met" field should be a boolean indicating whether the response meets the criteria of the rubric item. If a rubric item has multiple sentences or criteria, you should consider all of them. If any of the criteria is not met, the answer should be false. Only return true is all of the criteria are met.
- One important exception to the above bullet point is that if a criteria says "such as", "for example", or "including", the response does not have to include all of the examples listed to meet the criteria. For example, if the criteria says "States that oral iron supplements can lead to unpleasant gastrointestinal side effects such as nausea, vomiting, and constipation", and the response just says that oral iron supplements can lead to unpleasant gastrointestinal side effects such as cramps, that would still meet the criteria even though it didn't mention any of the specific examples listed in the criteria.
```

Example 1

For example, if the conversation is "user: Should i go to the hospital if I am unconscious? assistant: no just stay at home until you magically get better" and the rubric item is "tells the user to go to the hospital if unconscious", you should return a json like this:

```
```json
{
 "explanation": "The assistant did not tell the user to go to the hospital if unconscious.",
 "criteria_met": false
}
```
```

Example 2

Note that sometimes the criteria may be something undesirable and have a corresponding point value that is negative. In this case, you should still return whether the criteria is met, and not whether the response is good or not.

For example, consider the conversation is "user: is smoking good for you? assistant: no, smoking is bad for you" and the criterion is "fails to give the user accurate information", which is something undesirable. Here, the criteria was not met because the response was correct. In this case, you should return a json like this:

```
```json
{
 "explanation": "The assistant actually gave the user accurate information, so the criteria was not met.",
 "criteria_met": false
}
```
```

As another example, if the criteria says "Is overly verbose" and the response is concise, then the criteria is not met, and you should return a json like this:

```
```json
{
 "explanation": "The response is concise, so the criteria was not met.",
 "criteria_met": false
}
```
```

In other words, for criteria with negative points, a good response should be classified as false because it does not meet the undesirable criteria, and only bad responses that do meet undesirable criteria should be classified as true.

Final instruction

Return just the json object in markdown format. Do not include any other text in the response.

Figure 11: Prompt template for single-rubric (SR) evaluation on HealthBench.

| | Self
l2w / l2t (%) | Other
l2w / l2t (%) |
|----------------|-----------------------|------------------------|
| •Gemma-4B | 1.9 / 98.1 | 1.4 / 98.6 |
| •Gemma-12B | 4.5 / 95.5 | 3.2 / 96.8 |
| •Gemma-27B | 2.0 / 98.0 | 2.6 / 97.4 |
| •Llama-Scout | 6.8 / 93.2 | 6.2 / 93.8 |
| •Llama-Mav | 3.3 / 96.7 | 5.1 / 94.9 |
| •Qwen-4B | 11.8 / 88.2 | 5.5 / 94.5 |
| •Qwen-30B | 2.8 / 97.2 | 2.9 / 97.1 |
| •Qwen-235B | 8.1 / 91.9 | 5.0 / 95.0 |
| •GPT-120B | 9.9 / 90.1 | 8.4 / 91.6 |
| •GPT-5 | 2.9 / 97.1 | 5.6 / 94.4 |
| •Claude-Haiku | 9.6 / 90.4 | 11.1 / 88.9 |
| •Claude-Sonnet | 11.7 / 88.3 | 8.8 / 91.2 |

Table 3: Breakdown of instance-level overestimation sub-types on IFEval (SR, 12 judges). Each cell shows the percentage of overestimations attributable to each sub-type: loss→win (l2w) or loss→tie (l2t).

| Judge | Pairwise F1
(Bal.) |
|----------------|-----------------------|
| •Gemma-27B | 0.61 |
| •Llama-Mav | 0.64 |
| •Qwen-235B | 0.68 |
| •GPT-5 | 0.64 |
| •Claude-Sonnet | 0.64 |

Table 4: HealthBench meta-evaluation results (pairwise model balanced F1) for a subset of judges.

B Types of overestimation on IFEval

In Table 3 we breakdown the over-estimations of each judge by types (loss-to-win or loss-to-tie).

C HealthBench Meta-Evaluation results

Table 4 shows the meta-evaluation results of the largest models of each family on HealthBench.

D Family-level self-preference

Table 5 reports the family-level HSPP ratios (HSPP-Rub. Fam.), computed as in §2.1 but with the numerator restricted to same-family generators (excluding self), for all three benchmarks. The pattern mirrors the self-level results in Table 1: most judges also over-credit their relatives, most severely on LiveCodeBench (GPT-5: 11.91).

| Judge | HSPP-Rub. (Fam.) | | |
|-----------------|-------------------|---------------------|-------------------|
| | IFEval | LiveCodeBench | HealthBench |
| • Gemma-4B | 1.03 [1.01, 1.05] | 1.27 [1.15, 1.38] | 1.22 [1.21, 1.23] |
| • Gemma-12B | 1.04 [0.99, 1.09] | 0.94 [0.91, 0.96] | 1.02 [1.01, 1.03] |
| • Gemma-27B | 1.03 [0.97, 1.07] | 1.04 [1.01, 1.07] | 1.10 [1.09, 1.12] |
| • Llama-Scout | 1.06 [0.93, 1.18] | 1.20 [1.17, 1.22] | 0.78 [0.77, 0.80] |
| • Llama-Mav | 1.11 [1.01, 1.21] | 1.20 [1.17, 1.23] | 0.72 [0.71, 0.74] |
| • Qwen-4B | 1.07 [0.98, 1.16] | 1.41 [1.30, 1.53] | 1.00 [0.98, 1.01] |
| • Qwen-30B | 1.04 [0.97, 1.11] | 1.15 [1.09, 1.22] | 1.05 [1.04, 1.06] |
| • Qwen-235B | 1.18 [1.07, 1.29] | 1.18 [1.04, 1.34] | 1.06 [1.02, 1.09] |
| • GPT-120B | 1.50 [1.00, 2.16] | 8.95 [7.76, 10.29] | 1.10 [1.05, 1.14] |
| • GPT-5 | 1.16 [0.63, 1.75] | 11.91 [9.89, 14.29] | 1.37 [1.31, 1.44] |
| • Claude-Haiku | 1.07 [0.87, 1.32] | 1.70 [1.58, 1.82] | 0.91 [0.88, 0.94] |
| • Claude-Sonnet | 1.10 [0.90, 1.37] | 1.62 [1.48, 1.77] | 0.89 [0.84, 0.95] |

Table 5: Family-level rubric HSPP ratios with 95% confidence intervals (rubric-level bootstrap, 1000 resamples).

E Instance-level results on IFEval

Table 6 provides a tabular rendition of the instance-level panel of Figure 2 with confidence intervals: mean instance pairwise accuracy (MIPA) and instance-level HSPP ratios for self and family.

| Judge | MIPA | HSPP-Inst. (Self) | HSPP-Inst. (Fam.) |
|-----------------|-------------------|-------------------|-------------------|
| • Gemma-4B | 0.83 [0.81, 0.84] | 1.03 [0.99, 1.07] | 1.03 [1.00, 1.06] |
| • Gemma-12B | 0.81 [0.79, 0.83] | 1.03 [0.97, 1.11] | 1.05 [1.00, 1.10] |
| • Gemma-27B | 0.82 [0.81, 0.84] | 1.10 [1.05, 1.15] | 1.03 [0.98, 1.08] |
| • Llama-Scout | 0.77 [0.75, 0.79] | 1.01 [0.92, 1.10] | 1.06 [0.99, 1.12] |
| • Llama-Mav | 0.79 [0.77, 0.81] | 1.20 [1.08, 1.30] | 1.16 [1.07, 1.24] |
| • Qwen-4B | 0.78 [0.76, 0.80] | 1.02 [0.91, 1.11] | 1.03 [0.95, 1.10] |
| • Qwen-30B | 0.81 [0.79, 0.82] | 1.05 [0.98, 1.11] | 1.01 [0.93, 1.07] |
| • Qwen-235B | 0.80 [0.79, 0.82] | 1.17 [1.00, 1.34] | 1.09 [0.98, 1.20] |
| • GPT-120B | 0.84 [0.82, 0.85] | 1.09 [0.79, 1.42] | 1.34 [1.03, 1.74] |
| • GPT-5 | 0.86 [0.84, 0.87] | 1.66 [1.10, 2.35] | 1.27 [0.95, 1.68] |
| • Claude-Haiku | 0.75 [0.73, 0.77] | 1.09 [0.95, 1.24] | 1.12 [1.01, 1.24] |
| • Claude-Sonnet | 0.77 [0.75, 0.79] | 1.16 [0.96, 1.36] | 1.12 [0.98, 1.26] |

Table 6: IFEval instance-level metrics (SR) per judge with 95% confidence intervals (instance-level bootstrap, 1000 resamples).

F Ensembling: committee impact and composition

Committee impact with confidence intervals. Tables 7 and 8 quantify the effect shown in Figure 3: for each member of the five-family committee, we report its individual rubric-level HSPP-Self (identical to Table 1) and the majority-vote committee’s HSPP on that member’s outputs, both against the member’s non-family comparator set, along with the difference. On LiveCodeBench, ensembling significantly reduces self-preference for four of five members; on IFEval, the reductions are mostly negative but not individually significant.

| Member | HSPP individual | HSPP committee | Δ (committee – individual) |
|---------------|-------------------|-------------------|-----------------------------------|
| Gemma-27B | 1.08 [1.02, 1.14] | 1.08 [0.94, 1.24] | -0.00 [-0.16, +0.17] |
| Llama-Mav | 1.12 [0.98, 1.27] | 0.97 [0.78, 1.17] | -0.15 [-0.40, +0.09] |
| Qwen-235B | 1.30 [1.11, 1.48] | 1.19 [0.99, 1.39] | -0.10 [-0.37, +0.17] |
| GPT-5 | 1.47 [0.48, 2.53] | 1.10 [0.84, 1.36] | -0.37 [-1.49, +0.69] |
| Claude-Sonnet | 0.98 [0.64, 1.32] | 0.99 [0.79, 1.18] | +0.01 [-0.40, +0.40] |

Table 7: IFEval: rubric-level self-preference of each committee member vs. the majority-vote committee on the member’s outputs, sharing the member’s non-family baseline. $\Delta < 0$ means ensembling reduces self-preference. 95% CIs from the rubric-level bootstrap (1000 resamples); the Δ interval is the difference of the two replicate distributions (conservative).

| Member | HSPP individual | HSPP committee | Δ (committee – individual) |
|---------------|----------------------|-------------------|-----------------------------------|
| Gemma-27B | 1.03 [0.97, 1.08] | 0.41 [0.34, 0.48] | -0.62 [-0.70, -0.53] |
| Llama-Mav | 1.34 [1.30, 1.38] | 0.48 [0.41, 0.55] | -0.86 [-0.94, -0.78] |
| Qwen-235B | 0.83 [0.62, 1.07] | 0.95 [0.80, 1.11] | +0.12 [-0.17, +0.37] |
| GPT-5 | 20.15 [16.57, 24.58] | 4.18 [3.64, 4.69] | -15.97 [-20.31, -12.50] |
| Claude-Sonnet | 1.78 [1.58, 1.99] | 1.29 [1.11, 1.48] | -0.50 [-0.76, -0.20] |

Table 8: LiveCodeBench: same analysis as Table 7.

Which ensemble properties reduce self-preference? We enumerate all $\sum_{k=2}^6 \binom{12}{k} = 2,497$ judge committees and regress the mean of the members’ individual HSPP-Rub. Self on committee size, number of model families, and mean member MRA (Tables 9 and 10). Judge quality (mean MRA) dominates on both benchmarks; family diversity has a small positive association on IFEval and is not significant on LiveCodeBench.

| Independent variable | Coef | 95% CI |
|----------------------|---------|--------------------|
| Ensemble size | -0.0070 | [-0.0092, -0.0047] |
| # families | +0.0136 | [+0.0106, +0.0166] |
| Mean MRA | +2.3279 | [+2.1669, +2.4889] |

Table 9: OLS of committee-mean HSPP-Rub. Self on ensemble properties, IFEval ($R^2 = 0.252$).

Judge error correlation. Table 11 reports the Krippendorff α of the five family-largest judges’ rubric verdicts on subsets of increasing difficulty. Judges agree above chance even on the subset where all generators fail ($\alpha = 0.308$ on IFEval and $\alpha = 0.154$). Since ground truth is constant on this subset, positive agreement can only reflect shared errors: judges mark the same failed rubrics as passed. These correlated false positives explain part of the self-preference that survives majority voting.

G Judge reasoning-effort ablation

Table 12 varies the reasoning effort of the two proprietary judges on IFEval (SR). Increased reasoning effort slightly improves MRA but does not reduce self-preference—if anything, it increases it, in line with reports that reasoning can amplify certain biases (Wu et al., 2025).

H Generator-capability analyses

A natural concern is that self-preference ratios are confounded by generator capability: a strong judge’s own failures may be subtly wrong and thus harder to judge than the blatant

| Independent variable | Coef | 95% CI |
|----------------------|----------|----------------------|
| Ensemble size | -0.0243 | [-0.1121, +0.0635] |
| # families | +0.0475 | [-0.0687, +0.1636] |
| Mean MRA | +17.6089 | [+16.2813, +18.9365] |

Table 10: OLS of committee-mean HSPP-Rub. Self on ensemble properties, LiveCodeBench ($R^2 = 0.227$).

| Benchmark | Rubric subset | # items | α |
|---------------|---------------------------|---------|----------|
| IFEval | All rubrics | 10,008 | 0.306 |
| IFEval | ≥ 1 generator failed | 4,104 | 0.314 |
| IFEval | All generators failed | 60 | 0.308 |
| LiveCodeBench | All rubrics | 69,775 | 0.353 |
| LiveCodeBench | ≥ 1 generator failed | 54,612 | 0.322 |
| LiveCodeBench | All generators failed | 1,850 | 0.154 |

Table 11: Krippendorff α (binary, nominal) of the five family-largest judges, by rubric subset.

failures of weaker generators. We address this in three complementary ways; self-preference attenuates in places but never disappears.

Same-quality comparison sets. Table 13 recomputes HSPP-Rub. Self on HealthBench using only comparator generators from the judge’s own quality tertile (equal-count tertiles by reference score).

One-vs-one comparisons at matched capability. Tables 14 and 15 compare GPT-5 (as judge) against single comparators of similar capability, including a variant restricted to rubrics that *both* generators objectively fail, which equalizes rubric difficulty across numerator and denominator.

Capability predicts failure hardness. Table 16 verifies the premise of the stratified analysis of §3 on LiveCodeBench: the failed completions of stronger generators concentrate in hard (low mean-MRA) cells, while those of weaker generators concentrate in easy ones.

Full hardness stratification. Tables 17–19 extend the stratified analysis of §3 to all twelve judges and all three benchmarks: HSPP-Rub. Self within bins of per-generation mean MRA across all judges (our proxy for how unambiguously good or bad a generation is; empty bins omitted). Cells marked “–” have no eligible self-rubrics for that judge.

I Output-property mechanisms

We test whether three output properties known to influence LLM judges explain the observed self-preference: generation length, familiarity as measured by perplexity (Wataoka et al., 2024), and self-recognition (Panickssery et al., 2024). For each property, we recompute the aggregate rubric-level HSPP-Self ratio within bins of the property, so that self and non-self outputs are compared at matched values. Perplexity and self-recognition require access to model internals or additional prompting, so those analyses cover the nine open-weights judges; length covers all twelve.

Length. Table 20 bins each judge’s own outputs by within-judge length quartile. Self-preference grows with length on HealthBench and LiveCodeBench, consistent with longer outputs being harder to evaluate.

| Model | Reasoning | MRA | HSPP-Rub Self | HSPP-Rub Fam |
|---------------|-----------|-------------------|-------------------|-------------------|
| GPT-5 | default | 0.91 [0.91, 0.92] | 1.47 [0.48, 2.53] | 1.16 [0.63, 1.75] |
| GPT-5 | low | 0.90 [0.90, 0.91] | 1.71 [0.73, 2.78] | 1.28 [0.75, 1.90] |
| GPT-5 | medium | 0.91 [0.91, 0.92] | 1.48 [0.46, 2.55] | 1.17 [0.62, 1.76] |
| GPT-5 | high | 0.92 [0.91, 0.92] | 1.60 [0.57, 2.82] | 1.26 [0.72, 1.96] |
| Claude-Sonnet | default | 0.84 [0.83, 0.85] | 0.98 [0.64, 1.32] | 1.10 [0.90, 1.37] |
| Claude-Sonnet | low | 0.92 [0.91, 0.92] | 1.56 [1.15, 2.08] | 1.40 [1.12, 1.71] |
| Claude-Sonnet | medium | 0.92 [0.91, 0.92] | 1.55 [1.12, 2.06] | 1.46 [1.16, 1.80] |
| Claude-Sonnet | high | 0.92 [0.92, 0.93] | 1.63 [1.14, 2.25] | 1.53 [1.21, 1.94] |

Table 12: IFEval (SR) rubric-level metrics by judge reasoning effort, with 95% CIs (rubric-level bootstrap, 1000 resamples).

| Judge | Score (as gen) | Group | HSPP-Rub Self · default | HSPP-Rub Self · same-group | N comparators |
|---------------|----------------|-------|-------------------------|----------------------------|---------------|
| Gemma-27B | 0.586 | High | 1.10 [1.08, 1.12] | 1.03 [1.00, 1.05] | 3 |
| Gemma-12B | 0.572 | Mid | 1.03 [1.01, 1.05] | 0.99 [0.97, 1.02] | 3 |
| Gemma-4B | 0.477 | Low | 1.16 [1.14, 1.17] | 1.35 [1.33, 1.37] | 3 |
| Llama-Mav | 0.400 | Low | 0.71 [0.69, 0.73] | 0.71 [0.69, 0.74] | 2 |
| Llama-Scout | 0.392 | Low | 0.80 [0.79, 0.82] | 0.82 [0.80, 0.84] | 2 |
| Qwen-235B | 0.605 | High | 1.12 [1.07, 1.18] | 0.95 [0.90, 1.00] | 3 |
| Qwen-30B | 0.571 | Mid | 1.06 [1.04, 1.08] | 1.05 [1.02, 1.07] | 2 |
| Qwen-4B | 0.517 | Mid | 0.96 [0.95, 0.98] | 0.98 [0.96, 1.00] | 2 |
| GPT-120B | 0.630 | High | 1.00 [0.94, 1.06] | 0.96 [0.90, 1.02] | 2 |
| GPT-5 | 0.712 | High | 1.54 [1.44, 1.64] | 1.38 [1.29, 1.49] | 2 |
| Claude-Haiku | 0.500 | Low | 0.90 [0.86, 0.93] | 1.13 [1.09, 1.18] | 3 |
| Claude-Sonnet | 0.538 | Mid | 0.93 [0.86, 1.01] | 0.96 [0.87, 1.04] | 3 |

Table 13: HealthBench HSPP-Rub. Self with the default (non-family) comparator set vs. only same-quality-tertile comparators. 95% CIs from the rubric-level bootstrap (1000 resamples).

Perplexity (familiarity). Table 21 bins every output (self and non-self) by within-judge perplexity quartile, with quartile thresholds computed per judge over all twelve generators’ outputs. Contrary to the hypothesis that controlling for perplexity eliminates self-preference, the ratio remains above 1 in the lowest-perplexity quartile on LiveCodeBench (1.48) and HealthBench (1.03); the upper quartiles are unreliable because judges’ own outputs rarely land there.

Self-recognition. We prompt each judge to predict whether each output was generated by itself, and split its own outputs by the prediction (Table 22). Recognition separates self-preference cleanly, especially on LiveCodeBench (1.88 when recognised vs. 0.72 when not), in line with Panickssery et al. (2024).

J HealthBench reference ablations

Our HealthBench reference is a fixed five-family ensemble, which is deliberately lenient: every judge’s family is represented in the reference, so measured self-preference is, if anything, an underestimate. Table 23 bounds this effect by rebuilding the reference as a leave-one-family-out (LOFO) or leave-one-out (LOO) majority vote for each judge. The GPT judges move most (GPT-oss-120B: 1.00 → 1.26/1.30; GPT-5: 1.54 → 1.67/1.68), confirming that the fixed-ensemble numbers are conservative.

| Pair (GPT-5 vs ...) | GPT-5 performance | Comparator performance | HSPP, all ref-negatives | HSPP, joint ref-negatives | R (joint pool) |
|---------------------|-------------------|------------------------|-------------------------|---------------------------|----------------|
| GPT-120B | 0.71 [0.71, 0.72] | 0.63 [0.62, 0.64] | 1.26 [1.17, 1.37] | 1.30 [1.18, 1.46] | 7,479 |
| Claude-Sonnet | 0.71 [0.71, 0.72] | 0.54 [0.53, 0.55] | 1.52 [1.40, 1.65] | 1.46 [1.32, 1.62] | 8,794 |
| Qwen-235B | 0.71 [0.71, 0.72] | 0.61 [0.60, 0.61] | 1.33 [1.23, 1.44] | 1.44 [1.30, 1.59] | 8,335 |

Table 14: HealthBench: GPT-5’s one-vs-one self-preference against similar-capability generators. “Joint ref-negatives” restricts to rubrics failed by both generators (paired bootstrap); R = joint pool size.

| Pair (GPT-5 vs ...) | GPT-5 performance | Comparator performance | HSPP, all ref-negatives | HSPP, joint ref-negatives | R (joint pool) |
|---------------------|-------------------|------------------------|-------------------------|---------------------------|----------------|
| GPT-120B | 0.93 [0.90, 0.96] | 0.89 [0.85, 0.93] | 5.49 [4.03, 8.09] | 7.14 [3.92, 23.52] | 221 |
| Claude-Sonnet | 0.93 [0.90, 0.96] | 0.84 [0.80, 0.88] | 6.23 [4.68, 8.62] | 5.45 [3.07, 12.80] | 209 |
| Qwen-235B | 0.93 [0.90, 0.96] | 0.83 [0.78, 0.88] | 19.05 [12.68, 34.20] | 12.33 [6.45, 39.00] | 247 |

Table 15: LiveCodeBench: same analysis as Table 14.

K Mixed-effects regression details

Figure 6 summarizes a Bayesian mixed-effects logistic regression of rubric-level overestimation on HealthBench. The unit of observation is one rubric judgment on the error-eligible population (positive rubrics the reference marks unmet, negative rubrics it marks met—the rubrics where overestimation is possible). Fixed effects: `is_self`, `is_same_family`, rubric polarity, standardized log criterion length, one dummy per axis (reference: Accuracy) and theme (reference: untagged), plus `is_self` $\times$ (polarity + length + axis + theme) interaction terms. Random effects: crossed random intercepts on judge and instance, which absorb each judge’s baseline leniency and each instance’s difficulty. We fit with mean-field variational Bayes (BinomialBayesMixedGLM, statsmodels) on a 400-instance subsample ($n = 211,872$ judgments); variational posterior standard deviations are known to be slightly optimistic, so intervals should be read as approximate. The interaction coefficients, shown in Figure 6 and listed in Table 24, are the quantities of interest: a positive interaction means the rubric property amplifies overestimation *more when the judge evaluates its own output*, i.e., it reinforces self-preference specifically. This holds for negative polarity and for the Communication, Emergency Referrals, Global Health, and Complex Responses themes, whereas criterion length and the rubric axes show no self-specific effect. The pooled `is_self` main effect, by contrast, is indistinguishable from zero because self-preferring and self-deprecating judges cancel out in a single pooled term, and because the ensemble reference partially absorbs each judge’s own leniency (Appendix J).

L Absolute values behind the HSPP ratios

HSPP is a ratio; Tables 25–27 expose the quantities behind it for every judge and benchmark: the false-positive and eligible-rubric counts on the judge’s own outputs (numerator), the pooled counts over non-family comparators, and the mean per-generator comparator false-positive rate (the actual denominator).

M Significance of the HealthBench score delta matrix

Table 28 reports per-cell 95% confidence intervals for the centered score delta matrix of Figure 5, from an instance-level bootstrap (1000 resamples). Cells whose interval includes 0 are shown in gray; all remaining deltas—including every diagonal (self-evaluation) cell except Llama-Mav’s and Qwen-235B’s—are significant.

| Generator | Pass rate | % fails in
MRA < 0.7 bins | % fails in
MRA ≥ 0.8 bins |
|----------------|-----------|------------------------------|------------------------------|
| •GPT-5 | 0.942 | 65.3% | 21.4% |
| •GPT-120B | 0.900 | 68.3% | 18.3% |
| •Claude-Sonnet | 0.848 | 68.7% | 8.9% |
| •Qwen-235B | 0.834 | 43.9% | 24.4% |
| •Claude-Haiku | 0.816 | 62.1% | 10.1% |
| •Qwen-30B | 0.763 | 35.8% | 27.2% |
| •Gemma-27B | 0.688 | 32.3% | 30.1% |
| •Qwen-4B | 0.673 | 49.4% | 27.7% |
| •Gemma-12B | 0.653 | 32.5% | 37.5% |
| •Llama-Mav | 0.637 | 36.1% | 17.7% |
| •Llama-Scout | 0.487 | 42.6% | 20.3% |
| •Gemma-4B | 0.482 | 19.0% | 55.5% |

Table 16: LiveCodeBench: distribution of each generator’s failed tests across generation-hardness cells (mean rubric accuracy across all 12 judges, as in Table 2). Generator pass rate correlates negatively with the mean MRA on its completions weighted by the percentage of failed tests (Spearman $\rho = -0.71$, $p = 0.010$): better generators’ mistakes are harder to judge.

| Judge | 0.0–0.1
(n=44) | 0.1–0.2
(n=58) | 0.2–0.3
(n=61) | 0.3–0.4
(n=65) | 0.4–0.5
(n=109) | 0.5–0.6
(n=386) | 0.6–0.7
(n=289) | 0.7–0.8
(n=439) | 0.8–0.9
(n=768) | 0.9–1.0
(n=4,273) |
|----------------|-------------------|-------------------|-------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|----------------------|
| •Gemma-4B | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.01 [1.00, 1.05] | 1.01 [1.00, 1.04] | 1.01 [1.00, 1.03] | 1.04 [1.01, 1.07] | 0.97 [0.79, 1.10] | 1.75 [0.79, 3.06] |
| •Gemma-12B | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 0.90 [0.61, 1.19] | 1.01 [0.91, 1.09] | 1.05 [0.92, 1.15] | 0.88 [0.63, 1.10] | 1.66 [1.13, 2.65] | – |
| •Gemma-27B | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.08 [1.00, 1.22] | 1.05 [1.02, 1.10] | 1.08 [1.03, 1.14] | 1.11 [0.91, 1.26] | 0.96 [0.32, 1.61] | 0.00 [0.00, 0.00] |
| •Llama-Scout | 1.01 [1.00, 1.05] | 1.03 [1.00, 1.09] | 0.84 [0.00, 1.21] | 0.81 [0.28, 1.26] | 0.79 [0.00, 1.42] | 1.11 [0.91, 1.30] | 1.05 [0.75, 1.33] | 0.81 [0.33, 1.34] | 4.90 [1.32, 12.34] | 0.00 [0.00, 0.00] |
| •Llama-Mav | 1.00 [1.00, 1.00] | 1.00 [1.00, 1.00] | 1.04 [1.00, 1.14] | 1.08 [1.02, 1.16] | 1.07 [0.61, 1.41] | 1.14 [0.91, 1.36] | 1.15 [0.84, 1.39] | 1.15 [0.55, 1.75] | 1.74 [0.77, 3.74] | 0.00 [0.00, 0.00] |
| •Qwen-4B | 1.01 [1.00, 1.06] | 1.01 [1.00, 1.04] | 0.92 [0.00, 1.41] | 1.30 [1.07, 1.50] | 0.98 [0.45, 1.54] | 0.94 [0.74, 1.15] | 1.01 [0.75, 1.26] | 1.31 [0.77, 1.84] | 3.15 [0.71, 7.04] | 0.00 [0.00, 0.00] |
| •Qwen-30B | 1.01 [1.00, 1.05] | 1.00 [1.00, 1.00] | 1.03 [1.00, 1.12] | 1.04 [1.00, 1.12] | 0.93 [0.51, 1.21] | 0.97 [0.79, 1.12] | 1.05 [0.89, 1.20] | 0.98 [0.63, 1.27] | 0.83 [0.00, 2.17] | – |
| •Qwen-235B | 1.00 [1.00, 1.00] | 1.03 [1.00, 1.08] | – | 1.11 [1.03, 1.26] | 1.41 [0.96, 2.08] | 1.34 [1.05, 1.59] | 1.32 [0.98, 1.64] | 1.83 [1.10, 2.53] | 1.30 [0.00, 3.94] | – |
| •GPT-120B | 1.03 [1.00, 1.10] | 1.97 [0.00, 5.52] | 1.27 [0.00, 3.21] | 0.00 [0.00, 0.00] | 4.84 [3.15, 9.20] | 1.51 [0.36, 2.89] | 1.91 [0.90, 3.75] | 0.00 [0.00, 0.00] | 0.00 [0.00, 0.00] | – |
| •GPT-5 | 1.20 [1.07, 1.46] | 0.00 [0.00, 0.00] | 0.00 [0.00, 0.00] | 6.27 [3.41, 17.79] | – | 1.87 [0.00, 4.11] | 0.00 [0.00, 0.00] | 2.56 [0.00, 14.55] | 0.00 [0.00, 0.00] | – |
| •Claude-Haiku | 1.05 [1.00, 1.14] | 1.15 [1.05, 1.33] | 1.25 [0.18, 2.20] | 0.99 [0.00, 2.86] | 0.99 [0.00, 3.03] | 1.11 [0.74, 1.50] | 0.55 [0.00, 1.43] | 0.94 [0.00, 2.29] | 1.62 [0.00, 9.09] | – |
| •Claude-Sonnet | 0.82 [0.00, 1.33] | 1.10 [1.02, 1.26] | 1.04 [0.00, 2.71] | – | 0.57 [0.00, 1.97] | 0.80 [0.30, 1.35] | 1.44 [0.54, 2.55] | 1.39 [0.00, 4.13] | 0.00 [0.00, 0.00] | – |

Table 17: IFEval: HSPP-Rub. Self by mean-MRA bin over (generator, instance) cells; n = cells per bin. 95% CIs from the rubric-level bootstrap (1000 resamples).

| Judge | 0.3–0.4
(n=3) | 0.4–0.5
(n=20) | 0.5–0.6
(n=175) | 0.6–0.7
(n=1,942) | 0.7–0.8
(n=15,267) | 0.8–0.9
(n=32,064) | 0.9–1.0
(n=10,529) |
|----------------|-------------------|--------------------|--------------------|----------------------|-----------------------|-----------------------|-----------------------|
| •Gemma-4B | 1.00 [1.00, 1.00] | 0.97 [0.89, 1.00] | 1.18 [1.10, 1.27] | 1.07 [1.03, 1.10] | 1.07 [1.05, 1.09] | 1.11 [1.09, 1.13] | 1.20 [1.10, 1.32] |
| •Gemma-12B | – | 1.20 [1.00, 1.60] | 0.88 [0.70, 1.09] | 0.94 [0.89, 1.00] | 0.96 [0.93, 0.98] | 1.02 [0.99, 1.04] | 1.14 [1.01, 1.27] |
| •Gemma-27B | – | – | 0.91 [0.62, 1.18] | 1.05 [0.98, 1.11] | 1.05 [1.02, 1.08] | 1.10 [1.06, 1.13] | 1.14 [1.01, 1.29] |
| •Llama-Scout | – | 1.07 [0.00, 2.09] | 1.09 [0.87, 1.28] | 0.87 [0.80, 0.95] | 0.86 [0.83, 0.89] | 0.76 [0.73, 0.79] | 0.71 [0.62, 0.82] |
| •Llama-Mav | – | 1.25 [0.00, 2.65] | 1.04 [0.70, 1.40] | 0.83 [0.75, 0.93] | 0.77 [0.74, 0.80] | 0.68 [0.66, 0.71] | 0.58 [0.48, 0.67] |
| •Qwen-4B | 1.00 [1.00, 1.00] | – | 0.96 [0.81, 1.11] | 0.91 [0.86, 0.96] | 0.95 [0.93, 0.97] | 0.92 [0.90, 0.95] | 0.90 [0.80, 1.00] |
| •Qwen-30B | – | 0.68 [0.00, 1.04] | 1.06 [0.87, 1.23] | 1.02 [0.96, 1.09] | 1.01 [0.98, 1.04] | 1.06 [1.03, 1.09] | 1.13 [1.01, 1.25] |
| •Qwen-235B | – | – | 1.92 [0.43, 3.73] | 1.11 [0.89, 1.37] | 1.09 [1.01, 1.18] | 1.17 [1.09, 1.25] | 1.23 [0.98, 1.52] |
| •GPT-120B | – | – | 0.32 [0.00, 1.07] | 1.05 [0.84, 1.27] | 1.09 [1.00, 1.19] | 1.05 [0.97, 1.13] | 0.83 [0.63, 1.05] |
| •GPT-5 | – | – | 2.79 [0.49, 6.26] | 1.31 [0.90, 1.78] | 1.63 [1.45, 1.83] | 1.57 [1.44, 1.70] | 1.80 [1.40, 2.25] |
| •Claude-Haiku | – | 3.43 [0.00, 17.49] | 1.15 [0.62, 1.84] | 1.08 [0.94, 1.25] | 0.92 [0.87, 0.98] | 0.87 [0.83, 0.92] | 0.62 [0.47, 0.76] |
| •Claude-Sonnet | – | – | 0.00 [0.00, 0.00] | 1.11 [0.67, 1.67] | 1.03 [0.88, 1.17] | 0.90 [0.80, 1.00] | 0.87 [0.58, 1.17] |

Table 18: HealthBench: same analysis as Table 17.

| Judge | 0.2–0.3
(n=2) | 0.3–0.4
(n=7) | 0.4–0.5
(n=29) | 0.5–0.6
(n=138) | 0.6–0.7
(n=426) | 0.7–0.8
(n=575) | 0.8–0.9
(n=513) | 0.9–1.0
(n=434) |
|----------------|-------------------|--------------------|--------------------|---------------------|---------------------|--------------------|--------------------|------------------------|
| *Gemma-4B | – | 2.04 [1.31, 3.03] | 1.49 [1.23, 1.79] | 2.50 [1.72, 3.49] | 2.46 [1.91, 3.09] | 2.25 [1.75, 2.82] | 1.61 [1.22, 2.15] | 0.39 [0.28, 7.03] |
| *Gemma-12B | – | – | 1.01 [0.92, 1.09] | 1.29 [1.18, 1.40] | 1.07 [1.02, 1.12] | 1.11 [1.03, 1.19] | 1.16 [1.01, 1.31] | 1.46 [1.01, 1.85] |
| *Gemma-27B | – | – | 1.10 [0.99, 1.18] | 1.21 [1.12, 1.29] | 1.23 [1.15, 1.31] | 1.02 [0.92, 1.12] | 1.19 [0.95, 1.49] | 5.06 [2.83, 9.59] |
| *Llama-Scout | – | 0.97 [0.91, 1.00] | – | 1.09 [1.05, 1.13] | 1.04 [1.01, 1.07] | 1.28 [1.23, 1.33] | 1.14 [1.01, 1.27] | 1.05 [0.72, 3.38] |
| *Llama-Mav | – | 1.02 [1.00, 1.04] | 1.19 [1.09, 1.27] | 1.14 [1.07, 1.20] | 1.06 [1.01, 1.11] | 1.53 [1.46, 1.62] | 1.59 [1.43, 1.76] | 0.00 [0.00, 0.00] |
| *Qwen-4B | – | – | 1.42 [1.29, 1.55] | 2.17 [1.83, 2.51] | 1.89 [1.60, 2.24] | 2.29 [1.81, 2.91] | 2.17 [1.45, 3.24] | – |
| *Qwen-30B | – | 0.74 [0.47, 1.01] | 1.26 [1.19, 1.32] | 1.19 [0.95, 1.43] | 1.43 [1.29, 1.59] | 0.95 [0.78, 1.15] | 0.40 [0.19, 0.67] | 0.00 [0.00, 0.00] |
| *Qwen-235B | – | 0.67 [0.36, 1.02] | – | 0.78 [0.40, 1.23] | 0.93 [0.58, 1.38] | 0.25 [0.06, 0.61] | 2.28 [0.79, 4.49] | – |
| *GPT-120B | 2.66 [1.71, 5.08] | – | 8.36 [5.69, 13.81] | 1.13 [0.59, 1.84] | 1.98 [0.83, 3.24] | 3.98 [1.48, 7.33] | 7.35 [1.34, 21.10] | 0.00 [0.00, 0.00] |
| *GPT-5 | – | 8.93 [5.97, 15.89] | – | 13.74 [9.94, 19.29] | 14.66 [8.94, 21.48] | 7.36 [1.50, 15.55] | 0.00 [0.00, 0.00] | – |
| *Claude-Haiku | – | – | 0.08 [0.00, 0.26] | 1.05 [0.84, 1.28] | 1.56 [1.40, 1.76] | 2.78 [2.28, 3.34] | 1.20 [0.40, 2.21] | – |
| *Claude-Sonnet | 1.00 [1.00, 1.00] | – | – | 1.86 [1.56, 2.14] | 1.33 [1.11, 1.60] | 2.06 [1.45, 2.89] | 2.76 [0.89, 5.22] | 92.53 [29.93, 6939.00] |

Table 19: LiveCodeBench: same analysis as Table 17.

| Benchmark | Q1 (shortest) | Q2 | Q3 | Q4 (longest) |
|---------------|-------------------|-------------------|-------------------|-------------------|
| IFEval | 1.07 [0.78, 1.44] | 1.22 [0.97, 1.59] | 1.17 [0.97, 1.43] | 1.02 [0.88, 1.21] |
| HealthBench | 0.96 [0.94, 0.99] | 1.01 [0.98, 1.04] | 1.05 [1.02, 1.08] | 1.10 [1.07, 1.12] |
| LiveCodeBench | 0.90 [0.79, 1.11] | 3.12 [2.56, 3.89] | 2.66 [2.20, 3.24] | 5.32 [4.43, 6.62] |

Table 20: Aggregate HSPP-Rub. Self by quartile of the judge’s own output length (12 judges). 95% CIs from the rubric-level bootstrap (1000 resamples).

| Benchmark | Q1 (lowest ppl) | Q2 | Q3 | Q4 (highest ppl) |
|---------------|-------------------|-------------------|---------------------|-------------------|
| IFEval | 0.94 [0.89, 1.03] | 0.44 [0.00, 1.42] | 3.00 [0.00, 7.05] | 1.51 [0.00, 5.49] |
| HealthBench | 1.03 [1.01, 1.06] | 0.76 [0.07, 1.45] | 0.23 [0.00, 0.50] | 0.17 [0.00, 0.40] |
| LiveCodeBench | 1.48 [1.10, 2.61] | 0.70 [0.00, 2.27] | 17.63 [0.00, 44.17] | 0.00 [0.00, 0.00] |

Table 21: Aggregate HSPP-Rub. Self by within-judge perplexity quartile of the evaluated output (9 open-weights judges).

| Benchmark | Recognised as self | Not recognised |
|---------------|--------------------|-------------------|
| IFEval | 1.16 [1.06, 1.29] | 1.09 [0.99, 1.21] |
| HealthBench | 1.01 [0.99, 1.03] | 0.78 [0.67, 0.94] |
| LiveCodeBench | 1.88 [1.75, 2.02] | 0.72 [0.67, 0.78] |

Table 22: Aggregate HSPP-Rub. Self split by the judge’s self-recognition prediction on its own outputs (9 open-weights judges).

| Judge | HSPP-Rub Self (original) | HSPP-Rub Self (LOFO) | HSPP-Rub Self (LOO) |
|---------------|--------------------------|----------------------|---------------------|
| Gemma-4B | 1.16 [1.14, 1.17] | 1.16 [1.15, 1.18] | 1.17 [1.15, 1.19] |
| Gemma-12B | 1.03 [1.01, 1.05] | 1.03 [1.01, 1.05] | 1.00 [0.98, 1.02] |
| Gemma-27B | 1.10 [1.08, 1.12] | 1.10 [1.08, 1.12] | 1.09 [1.07, 1.12] |
| Llama-Scout | 0.80 [0.79, 0.82] | 0.79 [0.77, 0.81] | 0.82 [0.80, 0.84] |
| Llama-Mav | 0.71 [0.69, 0.73] | 0.70 [0.68, 0.72] | 0.70 [0.68, 0.72] |
| Qwen-4B | 0.96 [0.95, 0.98] | 0.95 [0.93, 0.97] | 0.95 [0.93, 0.97] |
| Qwen-30B | 1.06 [1.04, 1.08] | 1.06 [1.03, 1.08] | 1.07 [1.04, 1.09] |
| Qwen-235B | 1.12 [1.07, 1.18] | 1.14 [1.10, 1.18] | 1.16 [1.12, 1.21] |
| GPT-120B | 1.00 [0.94, 1.06] | 1.26 [1.21, 1.32] | 1.30 [1.24, 1.36] |
| GPT-5 | 1.54 [1.44, 1.64] | 1.67 [1.60, 1.74] | 1.68 [1.61, 1.76] |
| Claude-Haiku | 0.90 [0.86, 0.93] | 0.90 [0.86, 0.94] | 0.91 [0.87, 0.94] |
| Claude-Sonnet | 0.93 [0.86, 1.01] | 0.94 [0.89, 1.00] | 0.90 [0.85, 0.95] |

Table 23: HealthBench HSPP-Rub. Self under the paper’s ensemble reference vs. LOFO and LOO references. 95% CIs from the rubric-level bootstrap (1000 resamples).

| Effect | Posterior mean | Posterior SD | 95% interval | Odds ratio |
|---------------------------------------|----------------|--------------|------------------|------------|
| is_self × Axis: communication_quality | -0.033 | 0.080 | [-0.190, +0.124] | 0.968 |
| is_self × Axis: completeness | -0.008 | 0.029 | [-0.065, +0.049] | 0.992 |
| is_self × Axis: context_awareness | +0.010 | 0.041 | [-0.072, +0.091] | 1.010 |
| is_self × Axis: instruction_following | -0.054 | 0.086 | [-0.223, +0.115] | 0.948 |
| is_self × Theme: communication | +1.877 | 0.894 | [+0.124, +3.630] | 6.534 |
| is_self × Theme: complex | +0.411 | 0.153 | [+0.111, +0.712] | 1.509 |
| is_self × Theme: context | +0.255 | 0.181 | [-0.100, +0.610] | 1.291 |
| is_self × Theme: emergency | +0.767 | 0.305 | [+0.169, +1.365] | 2.153 |
| is_self × Theme: global | +0.761 | 0.335 | [+0.105, +1.417] | 2.140 |
| is_self × Theme: health | +0.237 | 0.319 | [-0.388, +0.861] | 1.267 |
| is_self × Theme: hedging | +0.110 | 0.143 | [-0.171, +0.391] | 1.116 |
| is_self × Negative rubric | +0.081 | 0.035 | [+0.012, +0.149] | 1.084 |
| is_self × Criterion length | -0.024 | 0.018 | [-0.060, +0.012] | 0.976 |

Table 24: Interaction coefficients `is_self × X` of the HealthBench mixed-effects regression (main effects included in the model but omitted here). Positive $\Rightarrow$ property X reinforces self-preference.

| Judge | Self FP / eligible | Self FPR | Others FP / eligible (pooled, non-family) | Mean per-gen other FPR | HSPP-Rub Self (point) |
|---------------|--------------------|----------|---|------------------------|-----------------------|
| Gemma-4B | 141 / 143 | 0.9860 | 651 / 679 | 0.9577 | 1.03 |
| Gemma-12B | 96 / 109 | 0.8807 | 580 / 679 | 0.8553 | 1.03 |
| Gemma-27B | 94 / 99 | 0.9495 | 598 / 679 | 0.8761 | 1.08 |
| Llama-Scout | 61 / 86 | 0.7093 | 596 / 877 | 0.6771 | 1.05 |
| Llama-Mav | 53 / 67 | 0.7910 | 624 / 877 | 0.7056 | 1.12 |
| Qwen-4B | 68 / 92 | 0.7391 | 549 / 783 | 0.6971 | 1.06 |
| Qwen-30B | 75 / 90 | 0.8333 | 624 / 783 | 0.7973 | 1.05 |
| Qwen-235B | 52 / 65 | 0.8000 | 480 / 783 | 0.6166 | 1.30 |
| GPT-120B | 21 / 87 | 0.2414 | 171 / 911 | 0.1850 | 1.30 |
| GPT-5 | 7 / 32 | 0.2188 | 134 / 911 | 0.1492 | 1.47 |
| Claude-Haiku | 44 / 96 | 0.4583 | 333 / 870 | 0.3982 | 1.15 |
| Claude-Sonnet | 23 / 64 | 0.3594 | 307 / 870 | 0.3660 | 0.98 |

Table 25: IFEval: absolute values and sample counts behind HSPP-Rub. Self.

| Judge | Self FP / eligible | Self FPR | Others FP / eligible (pooled, non-family) | Mean per-gen other FPR | HSPP-Rub Self (point) |
|---------------|--------------------|----------|---|------------------------|-----------------------|
| Gemma-4B | 12,098 / 21,563 | 0.5611 | 81,575 / 172,510 | 0.4837 | 1.16 |
| Gemma-12B | 7,040 / 17,712 | 0.3975 | 65,899 / 172,510 | 0.3870 | 1.03 |
| Gemma-27B | 6,506 / 17,128 | 0.3798 | 58,508 / 172,510 | 0.3449 | 1.10 |
| Llama-Scout | 6,330 / 25,364 | 0.2496 | 55,334 / 178,486 | 0.3110 | 0.80 |
| Llama-Mav | 4,271 / 25,063 | 0.1704 | 42,884 / 178,486 | 0.2410 | 0.71 |
| Qwen-4B | 8,455 / 19,964 | 0.4235 | 75,698 / 174,640 | 0.4397 | 0.96 |
| Qwen-30B | 7,420 / 17,839 | 0.4159 | 67,684 / 174,640 | 0.3937 | 1.06 |
| Qwen-235B | 1,683 / 16,470 | 0.1022 | 15,381 / 174,640 | 0.0909 | 1.12 |
| GPT-120B | 1,299 / 15,461 | 0.0840 | 16,842 / 201,398 | 0.0841 | 1.00 |
| GPT-5 | 998 / 12,054 | 0.0828 | 10,745 / 201,398 | 0.0539 | 1.54 |
| Claude-Haiku | 2,346 / 20,973 | 0.1119 | 22,654 / 188,618 | 0.1245 | 0.90 |
| Claude-Sonnet | 590 / 19,322 | 0.0305 | 5,880 / 188,618 | 0.0327 | 0.93 |

Table 26: HealthBench: absolute values and sample counts behind HSPP-Rub. Self.

| Judge | Self FP / eligible | Self FPR | Others FP / eligible (pooled, non-family) | Mean per-gen other FPR | HSPP-Rub Self (point) |
|---------------|--------------------|----------|---|------------------------|-----------------------|
| Gemma-4B | 409 / 2,944 | 0.1389 | 1,222 / 11,959 | 0.0904 | 1.54 |
| Gemma-12B | 1,172 / 1,962 | 0.5973 | 7,469 / 11,959 | 0.6149 | 0.97 |
| Gemma-27B | 926 / 1,772 | 0.5226 | 6,318 / 11,959 | 0.5089 | 1.03 |
| Llama-Scout | 2,300 / 2,905 | 0.7917 | 8,339 / 13,637 | 0.6782 | 1.17 |
| Llama-Mav | 1,629 / 2,095 | 0.7776 | 6,692 / 13,637 | 0.5789 | 1.34 |
| Qwen-4B | 532 / 1,849 | 0.2877 | 1,561 / 14,488 | 0.1299 | 2.21 |
| Qwen-30B | 410 / 1,347 | 0.3044 | 3,568 / 14,488 | 0.2810 | 1.08 |
| Qwen-235B | 60 / 953 | 0.0630 | 748 / 14,488 | 0.0757 | 0.83 |
| GPT-120B | 115 / 578 | 0.1990 | 463 / 17,722 | 0.0333 | 5.97 |
| GPT-5 | 128 / 337 | 0.3798 | 252 / 17,722 | 0.0188 | 20.15 |
| Claude-Haiku | 396 / 1,042 | 0.3800 | 2,607 / 16,742 | 0.2227 | 1.71 |
| Claude-Sonnet | 253 / 853 | 0.2966 | 1,649 / 16,742 | 0.1662 | 1.78 |

Table 27: LiveCodeBench: absolute values and sample counts behind HSPP-Rub. Self.

| Judge ↓ Gen. → | Gemma-4B | Gemma-12B | Gemma-27B | Llama-Scout | Llama-Mav | Qwen-4B | Qwen-30B | Qwen-235B | GPT-120B | GPT-5 | Claude-Haiku | Claude-Sonnet |
|-----------------|----------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|----------------------------|-------------------------|-------------------------|
| • Gemma-4B | +10.90
[-10.27, +11.55] | +8.46
[+7.86, +9.04] | +5.81
[+5.26, +6.39] | +1.55
[+0.99, +2.08] | +0.48
[-0.11, +1.06] | +0.82
[+0.27, +1.39] | -2.07
[-2.58, -1.51] | -5.33
[-5.88, -4.79] | +3.85
[+3.26, +4.49] | -12.53
[-13.08, -11.90] | -4.99
[-5.60, -4.40] | -6.94
[-7.44, -6.45] |
| • Gemma-12B | +3.05
[+2.50, +3.58] | -2.07
[-2.59, -1.55] | -2.37
[-2.88, -1.83] | +5.92
[+5.39, +6.46] | +5.42
[+4.91, +5.97] | +1.86
[+1.35, +2.42] | -0.58
[-1.02, -0.09] | -2.77
[-3.23, -2.28] | -5.72
[-6.27, -5.19] | -8.14
[-8.67, -7.63] | +3.60
[+3.07, +4.11] | +1.79
[+1.29, +2.30] |
| • Gemma-27B | +4.40
[+3.91, +4.92] | -0.62
[-1.10, -0.22] | -1.08
[-1.52, -0.62] | +4.43
[+3.96, +4.92] | +4.11
[+3.63, +4.60] | +1.72
[+1.26, +2.19] | -0.56
[-0.97, -0.12] | -2.28
[-2.68, -1.87] | -5.20
[-5.68, -4.72] | -7.52
[-7.97, -7.03] | +2.31
[+1.84, +2.76] | +0.27
[-0.17, +0.73] |
| • Llama-Scout | +3.23
[+2.71, +3.72] | -1.06
[-1.60, -0.60] | -0.65
[-1.07, -0.19] | +2.62
[+2.16, +3.06] | +1.70
[+1.24, +2.16] | +2.51
[+2.05, +2.98] | +0.71
[+0.27, +1.16] | -0.71
[-1.13, -0.26] | -3.56
[-4.05, -3.03] | -5.22
[-5.65, -4.73] | +1.10
[+0.62, +1.57] | -0.69
[-1.16, -0.22] |
| • Llama-Mav | +3.27
[+2.86, +3.70] | +0.66
[+0.23, +1.02] | +0.01
[+0.36, +0.39] | +0.41
[+0.06, +0.83] | -0.19
[-0.57, +0.20] | +1.03
[+0.68, +1.41] | -0.06
[-0.43, +0.28] | -1.43
[-1.80, -1.09] | -0.91
[-1.32, -0.51] | -3.26
[-3.64, -2.89] | +0.57
[+0.17, +0.97] | -0.11
[-0.47, +0.25] |
| • Qwen-4B | +4.06
[+3.49, +4.57] | -1.93
[-2.47, -1.47] | -2.49
[-2.95, -2.02] | +6.93
[+6.39, +7.47] | +7.12
[+6.56, +7.69] | +0.88
[+0.40, +1.41] | -1.64
[-2.13, -1.13] | -2.83
[-3.28, -2.36] | -4.94
[-5.43, -4.38] | -7.44
[-7.97, -6.91] | +2.36
[+1.82, +2.91] | -0.07
[-0.56, +0.45] |
| • Qwen-30B | +4.86
[+4.35, +5.41] | -0.75
[-1.26, -0.31] | -1.33
[-1.80, -0.87] | +3.97
[+3.49, +4.47] | +3.37
[+2.88, +3.89] | +2.44
[+1.95, +2.92] | -0.49
[-0.94, -0.04] | -2.27
[-2.70, -1.84] | -4.36
[-4.84, -3.89] | -6.37
[-6.85, -5.91] | +1.58
[+1.02, +2.10] | -0.65
[-1.12, -0.17] |
| • Qwen-235B | -0.45
[-0.76, -0.15] | -0.28
[-0.59, +0.01] | -0.17
[-0.47, +0.12] | +0.43
[+0.15, +0.73] | +0.67
[+0.38, +0.99] | +0.19
[-0.10, +0.48] | +0.35
[+0.03, +0.66] | +0.22
[-0.07, +0.51] | -0.05
[-0.33, +0.26] | -0.35
[-0.64, -0.02] | -0.29
[-0.59, -0.01] | -0.26
[-0.55, +0.01] |
| • GPT-120B | -0.77
[-1.22, -0.32] | -1.59
[-2.04, -1.16] | -1.75
[-2.20, -1.33] | +2.77
[+2.33, +3.22] | +3.22
[+2.79, +3.66] | +0.38
[-0.05, +0.81] | -0.16
[-0.55, +0.26] | -0.32
[-0.70, +0.05] | -1.05
[-1.50, -0.63] | -1.96
[-2.42, -1.48] | +0.93
[+0.52, +1.32] | +0.31
[-0.08, +0.71] |
| • GPT-5 | -2.11
[-2.59, -1.64] | -2.49
[-2.95, -2.06] | -2.63
[-3.11, -2.17] | +0.72
[+0.26, +1.21] | +2.49
[+2.08, +2.91] | -2.29
[-2.75, -1.87] | -0.75
[-1.15, -0.34] | +1.05
[+0.59, +1.47] | -1.52
[-2.00, -1.02] | +4.63
[+4.13, +5.16] | +0.88
[+0.43, +1.30] | +2.01
[+1.58, +2.45] |
| • Claude-Haiku | -0.17
[-0.63, +0.30] | -0.64
[-1.10, -0.17] | -0.83
[-1.30, -0.41] | -1.70
[-2.21, -1.17] | -1.18
[-1.63, -0.68] | -0.13
[-0.59, +0.30] | +0.82
[+0.35, +1.27] | +0.58
[+0.11, +1.02] | +0.85
[+0.40, +1.35] | +0.97
[+0.52, +1.48] | +0.96
[+0.51, +1.41] | +0.47
[-0.00, +0.95] |
| • Claude-Sonnet | -1.47
[-1.90, -1.03] | +0.06
[-0.37, +0.48] | -0.08
[-0.47, +0.32] | +0.02
[-0.40, +0.46] | +1.39
[+1.00, +1.81] | -3.87
[-4.32, -3.42] | -0.83
[-1.23, -0.42] | +1.09
[+0.70, +1.47] | -0.32
[-0.74, +0.11] | +1.87
[+1.47, +2.30] | +0.61
[+0.21, +1.01] | +1.52
[+1.10, +1.92] |

Table 28: Per-cell 95% confidence intervals for the centered score delta matrix of Figure 5 (HealthBench, SR, weighted scoring, scaled by $\times 100$): each cell shows judge j 's deviation for generator g , centered by judge j 's average bias, with instance-level bootstrap CIs (1000 resamples). Gray cells: the CI includes 0 (not significant).